AI Trend Detection in Healthcare by applying Topic Clustering and Sentiment Analysis using Podcast Data

Master's Thesis in Computer Science

submitted
by

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Übersicht

Künstliche Intelligenz (KI) wirkt sich auf die Entwicklung neuer Technologien in vielen Branchen aus. Die Auswirkungen auf das Gesundheitswesen sind von besonderem Interesse, da KI direkten Einfluss auf den gesundheitlichen Zustand von Patienten haben kann oder sogar Leben rettet. Daher ist es von entscheidender Bedeutung zu wissen, was im medizinischen Kontext diskutiert wird und welche Themen im Trend liegend. Der Podcast ist ein aufstrebendes Medium, das seit 2015 viel Aufmerksamkeit und aktive Zuhörer gewonnen hat. Podcasts über das Gesundheitswesen mit Gastrednern aus diversen Forschungsbereichen wie Medizin, Informatik oder Robotik könnten eine nützliche Quelle sein, um aktuelle Trends der KI im Gesundheitswesen besser zu identifizieren. Untersuchungen zur Trenderkennung von KI-bezogenen Themen in Medien wie Zeitungen oder Regierungsdokumenten haben die Effektivität von Trendanalysen zu KI-Themen bereits mehrfach bewiesen. Eine ausführliche Literaturrecherche ergab jedoch, dass das Podcasts im Zusammenhang mit KI-Trenderkennungen nicht ausreichend genug erforscht sind. Daher wird in dieser Arbeit die Eignung des Mediums Podcast für die Durchführung von KI-bezogener Trenderkennung untersucht. Aufgrund des Audioformats von Podcasts wurde in einem umfassenden Ansatz Vosk als die beste Open-Source-API für die Sprache-zu-Text-Transkription von Podcasts identifiziert. In weiteren Forschungsarbeiten wird untersucht, ob eine Themenclusterung und Stimmungsanalyse zur Erkennung zukünftiger KI-Trends verwendet werden können. In dieser Arbeit wird eine Pipeline vorgestellt, die eine Liste aus 102 Schlüsselwörtern und die Vosk-Audiotranskriptionen für 29 Podcasts aus dem Gesundheitswesen mit 3.449 Episoden nutzt. Aus allen Episoden und der Schlüsselwortliste wurde ein Datensatz auf Schlüsselwortebene mit 35.816 Textbausteinen erstellt. Es wurde eine Cosinus-Ähnlichkeits- und Co-Occurrence-Matrix-Analyse von 2015 bis 2021 durchgeführt. Es wurden Trendverläufe von vierzehn gesundheitsbezogene Themen identifiziert, wie z. B. Cloud Architecture, Healthcare Innovation oder KI. Eine Stimmungsanalyse wurde zusätzlich von 2015 bis 2021 zu den vierzehn gefundenen Themen berechnet. Zusammenfassend lässt sich sagen, dass eine Themenclusterung und Stimmungsanalyse praktische Werkzeuge sind, um Themenentwicklungen über einen vordefinierten Zeitraum zu identifizieren. Die im Rahmen dieser Arbeit entwickelten Technologien können auch Podcasts und deren Episoden nach den gewünschten Themen wie KI oder Cloud-Architektur filtern.

Abstract

Artificial intelligence impacts the development of new technologies in many industries. The impact in the healthcare sector is of special interest, because AI directly influences the well-being of patients or even can save lives. Therefore, knowing what is being discussed in the medical community and which topics are trending is crucial. Podcast are an emerging medium that have gained much attention and active listeners from 2015 onwards. Healthcare podcasts with guest speakers from different backgrounds, such as medicine, computer science or robotics could be a useful source to identify current trends of AI in the healthcare sector. Research about trend detection of AI-related topics in media like newspapers or governmental documents demonstrated the effectiveness of trend analysis about AI topics. However, an extended literature review revealed that podcasts lack research in the context of AI trend detection. Therefore, this thesis examines the suitability of the medium podcast for performing AI-related trend detection. Due to the audio format of podcasts, a comprehensive approach identified Vosk as the best performing open-source API for podcast speech-to-text transcriptions. Additional research includes investigating whether topic clustering and sentiment analysis can be used to detect future AI-related trends. This work presents a pipeline that leverages a keyword list with 102 keywords and the Vosk audio transcriptions for 29 healthcare podcasts with 3,449 episodes. A keyword-level data set of 35,816 text chunks was built from all episodes and the keyword list. A cosine similarity and co-occurrence matrix analysis from 2015 until 2021 was performed. Fourteen healthcare-related topics were identified, from which some were trending at certain time periods, such as Cloud Architecture, Healthcare Innovation or AI. The sentiment analysis was disclosed from 2015 until 2021 towards the fourteen found topics. In summary, topic clustering and sentiment analysis are practical tools to identify topic developments over a specified period, upcoming hypes, or declines of topics. The technology developed for this thesis can filter podcasts and episodes for the desired topics such as AI or cloud architecture.

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Chapter 1

Introduction

1.1 Motivation

The medium podcast has gained much attention and active listeners from the year 2015 onwards [Not; Kur21]. People from all demographics not only enjoy listening to podcasts, but podcasts also have a proven impact on forming listeners' opinions when podcast hosts invite guests to their shows with different perspectives and fields of expertise [Swi18]. Because there is a lack of research on trend detection with the medium podcast, it is crucial to find out what podcast hosts are talking about and how topics develop over a specified period of time. The research will be conducted with state-of-the-art data analysis techniques based on Deep Learning (DL).

This work builds upon an already conducted and validated prior work from Do and colleagues [Phi21], which investigated healthcare podcasts as a research medium and elaborated if the transcribed text from the podcasts can detect Artificial Intelligence (AI) related trends. The team identified the two best open-source technologies for performing speech-to-text transcriptions on the podcast audio files, namely Vosk [Alp] and DeepSpeech [Moz]. This thesis' goal is to elaborate on the currently best-performing technology for the transcription of podcast audio files by evaluating random audio snippets with the Word Error Rate (WER) metric [Kla00]. Furthermore, the already collected podcast data from 2010 to 2020 shall be extended by the year 2021 and advanced analysis using topic clustering and sentiment analysis to detect future trends will be conducted.

1.2 Structure of this Work

The three main topics that will be covered in this thesis can be formulated as the following research questions:

- What is the qualitative difference between the open-source speech-to-text APIs DeepSpeech and Vosk with respect to their WER and quantitative differences in the analysis of the data set?
- How suitable is topic clustering to identify keyword clusters that illustrate the relationships of the keywords for an AI trend analysis?
- What is the relationship between the speakers' sentiment and the topics found by topic clustering?

In Chapter 2, a literature research on podcasts as a research medium, podcasts for trend detection, topic clustering and sentiment analysis was conducted. The following Chapter 3 briefly introduces the main concepts and technologies that have been used throughout this thesis. Chapter 4 consists of multiple steps. First, the audio crawling pipeline, the speech-to-text Application Programming Interface (API) evaluation and the keyword list from Do et al. [Phi21] are presented. As a second step, the building process of the keyword-level data set based on the transcriptions and the keyword list is described. Topic clustering and sentiment analysis were finally implemented and fed with the keyword-level data set to perform trend detection. The results in the form of co-occurrence matrices and cosine similarity heatmaps from 2015 until 2021 are visualized and described in Chapter 5. The interpretations to see if future trend prediction of AI in healthcare podcasts is possible are further discussed in Chapter 6. The conclusion, possible limitations of the used methods and the future outlook of what could be improved or tested are provided in Chapter 7.

Chapter 2

Related Work

2.1 Podcast as a Research Medium

Bottomley [Bot15] investigated podcasts from 2005 to 2015 to find out if podcasts are a suitable and novel research medium. He conducted literature research on the medium podcast and correctly predicted that its popularity would rise in the future. His work was published in 2015. At this time, the noticeable rise of podcasts popularity has just begun in 2015 [Lis] (see Figure 5.1).

A team from Spotify published the first large-scale data set for free research usage. This data set contains 100,000 podcast episodes that have been randomly sampled between the years 2019 and 2020 from the Spotify platform. All podcasts combined have a total length of 60,000 hours of speech. The researchers concluded that podcasts could serve as an exciting novel research medium and provided it to the public [Cli20].

Do et al. [Phi21] investigated if the medium podcast is a suitable research medium to detect AI-related trends in podcasts of the genre healthcare. The team identified 30 suitable healthcare podcasts that started recording between 2015 and 2021. A trend analysis was performed by searching for predefined AI-related buzzwords in the over 3,000 crawled and transcribed podcast episodes. This approach was performed in the design of an environmental scanning approach, which is a standard procedure in the context of trend analysis in social sciences [Liu09]. Do showed, that podcasts are indeed an exciting and novel research medium, especially in detecting past trends of AI in healthcare podcasts. Their research showed, that there are still many open questions. More in-depth analysis with different tools, such as topic

clustering and sentiment analysis, could reveal more insights about future AI-related trends. It is tough to transcribe continuous speech with many word repetitions and speech stutterings correctly, which is the nature of podcast speech [Fet21]. Researchers tried to build language models specifically designed to tackle these problems in podcast speech in 2009 [Oga09].

2.2 Trend Detection

Becker et al. [Bec11] collected over 2.6 million Twitter posts in February 2010 from people who set their home town to be New York City on Twitter. The researchers wanted to identify real-world trends, such as presidential inauguration or high-school homecoming, by training a classifier to predict if a Twiiter hype was happening in real-time or if it was just hyped by Twitter users and actually did not happen at all or at a different time. The team used human annotation to have ground-truth even labels for the Twitter data. A classifier was trained on the data and evaluated with the F1-Score. The research has proven to be a successful end-to-end approach to identifying real-world events with Twitter content.
Even though podcasts were already identified as an emerging research medium in the year 2015 by Bottomley [Bot15], there is a lack of research on using the medium podcast for trend detection. Podcasts do not have a clear structure like newspaper articles, making it harder to process and clean the text. Nevertheless, the sheer amount of available podcast data makes it a rich and exciting new data source.
Welz and colleagues [Wel21] investigated if podcasts are a suitable medium for trend detection. The research group found out, that trend detection of specified topics can be detected correctly. Nevertheless, they also stated that it is necessary to identify a predefined list of potentially suitable podcasts according to the scanning algorithm for trend detection. This finding validates the approach from Do and his team [Phi21], which came to the same conclusion by selecting 30 podcasts that talk about AI and emerging technology-related topics beforehand.

2.3 Topic Clustering and Sentiment Analysis

Fast and Horvitz [Fas17] investigated if New York Times articles from 1986 to 2015 could identify AI-related trends within this time horizon. The team used crowd-sourcing to annotate paragraphs of the articles with adjectives like engagement, optimism vs. pessimism or hopes

and concerns for AI. This approach can be viewed as an extension of the sentiment analysis, because it includes both sentiment and emotions. Fast and Horvitz showed, that since 2009, AI has steadily gained popularity and AI-related buzzwords were frequently mentioned in New York Times articles. Their trend detection approach revealed an increased fear of AI and a fear of losing control. With structured and annotated data, trend prediction of AI-related topics towards certain emotions is possible.

Aghababaei et al. [Agh18] predicted crimes in certain parts of Chicago by crawling Twitter posts from people living in Chicago between 2010 and 2013 as well as local crime indices from official governmental homepages. For this purpose, features were extracted from the Twitter posts to train a topic model from the years 2010 to 2013 to visualize changes of topics over time. The basis of the temporal model is a regression model that predicts continuous local crime rates with official crime indices as ground-truth labels. Their research proved that temporal features extracted with topic clustering and sentiment analysis at a predefined frequency, in this case daily, can successfully capture trend developments.

The research group from Spotify [Cli20] not only created and provided the largest database with 100,000 podcast episodes from 2019 to 2020 to the public, but they also analyzed the data to find topics in the transcriptions with a predefined search query. The team concluded, that more sophisticated analysis techniques like topic clustering and sentiment analysis could improve the query search and identify more diverse topics. They explicitly mentioned topic clustering and sentiment analysis as future work, because those two approaches extend their previous work.

Do and his team [Phi21] also came to the same conclusion and suggested investigating further topic clustering and sentiment analysis for the podcast transcriptions. Therefore, these two approaches will be elaborated on in this thesis as primary methods for the trend analysis.

The related work from podcast as a research medium has shown that in 2006 podcasts were identified to be novel and their growth was already predicted due to their easy distribution through RSS. Nevertheless, the domain-specific rise of healthcare podcasts has increased significantly from 2015 on (Figure 5.1). Researchers are trying already since two decades to accurately predict trends. The literature in this sector revealed that an increase in performance of trend detection arose within the last decade, also due to growing data, e.g., through Clifton et al. [Cli20]. It showed that topic clustering and sentiment analysis tools were not popular until 2016. An increase in topic clustering methods for social media data like Twitter has been shown in this literature review. However, no advanced trend detection with podcast data

has been elaborated yet. This yields back to the research questions for this thesis, to build a pilot model for podcast data to detect future trends using topic clustering and sentiment analysis. The extended literature review can be found in Table 2.1.

Topic	Author(s)	Data	Methodology	Key Findings
Podcast as a Research Medium	Johnson and Grayden (2006) [Joh06]	Literature review	Research of how podcasts have originated, elaboration on Really Simple Syndication (RSS) Feeds	Distinction between amateur and professional podcasters, predict that podcast will grow substantially from 2006 on
	Bonini (2015) [Bon15]	Literature review from 2006 to 2014	Literature review about podcasts, literature review about broadcasting	The medium podcast is distinct from broadcasting, but podcasts nevertheless replaced broadcasts
	Berry et al. (2016) [Ber16]	Extended literature review from 2004 to 2016	Literature review about podcasts over the last decade	The medium podcast is distinct from other media like radio or YouTube and therefore requires novel research
Trend Detection	Cornwall (2007) [Cor07]	Review collection in *The Development Dictionary*	Buzzword Critique Development Discourse	Use of buzzwords to generalize and gain popularity, loss of subtle meaning
	Budak et al. (2011) [Bud11]	Twitter data (41.7 million nodes, 417 million posts)	Diffusion model, structural trend detection, social network analysis	Two new structural trend definitions, Use of friendship information for topic identification
	Caled et al. (2016) [Cal16]	DBLP database, 3 million bibliographic documents	Clustering, K-Means, self-organizing maps	Characteristics for buzzwords, connection between buzzwords and innovation
	Holzinger et al. (2018) [Hol18]	Tech sector, government funding, and AI conference attendance	Knowledge extraction, (statistical) ontologies	Mechanism in data-driven research, AI hype due to success in ML and knowledge extraction
Topic Clustering & Sentiment Analysis	Nikolenko et al. (2017) [Nik17]	235,407 Russian blog journals, dictionary with 700,000 tokens	Latent Dirichlet Analysis (LDA) topic clustering [Ble03], interval semi-supervised LDA	Topic clustering for qualitative studies lacks of good quality metrics
	Reagan et al. (2017) [Rea17]	6 sentiment dictionaries, 4 corpora (NYT [San08], Google Books [Lin12], Movie reviews [Pan04], Twitter)	Sentiment analysis, time-series analysis, Pearson correlation	Sentiment analysis has different challenges for different text data sources
	Zakkar and Lizotte (2021) [Zak21]	Healthcare experience data: 367,573 patient stories from 2005 to 2019	Text mining, sentiment analysis, LDA topic modeling	Sentiment analysis and topic modeling demonstrated to be good tools to identify critical issues in healthcare experiences

Table 2.1: Extended literature review on podcasts as a research medium, trend detection, topic clustering and sentiment analysis.

Chapter 3

Fundamentals

3.1 Podcast

A podcast is primarily a collection of individual media contributions, typically in the form of episodes. The term podcast is composed of *pod* from iPod (Apple's MP3 player) and *cast*, coming from the word broadcast [McG]. One or multiple hosts usually record podcast episodes that talk about specific subjects. The episodes are made publicly available through, e.g., the RSS-Feed with a link. A common way of listening to podcasts is via an app like Spotify [Spo] or Apple Podcast [App], but podcasts can also be subscribed for free via the RSS-Feed. This standard is mostly used for the distribution of information. RSS makes the access to podcast episodes very easy and allows users to download episodes with their corresponding metadata, like the publishing date or the names of guest speakers. The duration of one episode varies a lot, but most common is a length of up to an hour.

3.2 Automatic Speech Recognition

Automatic Speech Recognition (ASR) is used to transcribe speech into text automatically. Speech recognition consists of multiple building blocks [Yu15]:

- An acoustic model, which represents the relationship between audio signals and the building blocks of words,
- A language model, which connects the building blocks of words into words themselves,

- the phonetic dictionary, which models the decomposition of words into sounds [Alp].

Mature ASR systems, such as the Microsoft Azure speech-to-text API [Mic], have been trained on an extensive amount of audio. As described by Do and his colleagues [Phi21], these technologies tend to perform better for transcribing podcasts, but they also need to be purchased. However, good-performing free-to-use open-source solutions exist, which will be further evaluated in this thesis. A transcription is necessary to turn *MP3* or *WAV* files into *txt* files, which can then be used as input for various application fields in Natural Language Processing (NLP), such as topic clustering or text generation. The full functionality of an ASR system is very complex. First, the feature extraction is performed, using the raw audio signals and preprocessing them using normalization, windowing and a (log) spectrogram. Next, the acoustic model predicts the probability distributions of each character in a phonetic dictionary at each time step. Lastly, the decoder crafts the most likely text output for the acoustic model's probability distribution [Adr]. More details of how ASR systems work can be found in the automatic speech recognition book by Yu and colleagues [Yu15].

3.3 Natural Language Processing

NLP involves all tasks that operate on text. Language is transformed into bits of information, e.g., a numerical vector representation of the text. When dealing with text instead of numbers, Machine Learning (ML) and DL algorithms need a preprocessed version of the text, so they can use it to train models [Cho20]. NLP includes a bright variety of application fields, such as machine translation, question answering or topic clustering. Since this thesis examines a large amount of transcribed text from healthcare podcasts, a NLP pipeline is necessary. The pipeline involves the transformation of text into a vector representation and a cleaning pipeline to exclude words that are not necessary for the later model training and inference. A basic NLP pipeline depends on the task it is used for, but some standard operations are used most of the time. A standard topic clustering pipeline could consist of the following steps [Cho20]:

- Word tokenization (splitting senteces into single words),
- Lemmatization (reducing words to their word stems),
- Identifying stopwords (such as *and*, *or*),

- Transforming words into vector representation,
- Model training,
- Inference.

3.3.1 BERT

The Bidirectional Encoder Representations from Transformers (BERT) architecture was first introduced by Google in 2018 by Devlin [Dev18] and was built upon the Attention Network, also developed by the Google team [Vas17]. BERT is a neural network based on the so-called transformer architecture. This architecture consists of two main parts, an encoder and a decoder. The encoder uses text as input and the decoder outputs the text as a vector representation. This process can be viewed as feature extraction, because each inputted word will be transformed into a vector. These feature vectors are agnostic because they are not adapted to any specific task, but can be used as features for various NLP tasks, such as question answering or topic clustering [Rav21]. The BERT architecture consists of many layers of other neural network architectures, like the attention network [Vas17] or feed-forward neural networks. In summary, BERT is state-of-the-art for text feature extraction [Rav21]. Therefore, it will be considered the basic model for the topic clustering and the sentiment analysis used in this thesis.

3.3.2 Open Source Transformer Fine-tuning

Due to the excellent performance of BERT [Dev18] and transformers in general, an open-source community evolved around fine-tuned transformer models. Transfer learning allows training a model on a large data set once, and re-using the first few layers of the trained network for other tasks [Wei16]. Since the weights of the neural network's first few layers are very similar for tasks within the same general domain, transfer learning re-uses these layers and fine-tunes only the last few layers on the desired output. Google's team trained BERT on Wikipedia texts, resulting in a vocabulary richness of around 30,000 words. The training was performed on Google's servers and took multiple days. It would be environmentally unfriendly if everyone trained their models from scratch, because of the tremendous amount of electricity that is need for doing the same kind of training over and over again. Furthermore,

a training process for large transfers like BERT is estimated to cost from multiple ten thousand to millions of Dollars [Sha20].
The Hugging Face team [Tho20] identified this issue and created an open-source platform that enables an easy upload and distribution of already pre-trained transformer models. State-of-the-art transformers like BERT [Dev18], A Lite BERT for Self-supervised Learning (ALBERT) [Lan19] or the more novel Cross-lingual Language Model Pretraining (XLM) [Lam19] architecture are provided on the Hugging Face platform as well as models that have been fine-tuned by users. Each fine-tuned model stems from a transformer model that has been fine-tuned to solve a domain-specific task such as topic clustering, sentiment analysis or machine translation. For this reason, fine-tuned models that have been trained on BERT work only in the English language. If other languages such as English are required, a fine-tuned model based on a BERT model trained on a different language is necessary. If such a fine-tuned model is not made available by a user, it must be trained from scratch. Ideally, the newly fine-tuned model owner provides it publicly on the Hugging Face platform.

3.4 Topic Clustering

3.4.1 Algorithms

Topic clusters are content organizations that group content into clusters surrounding the main topic. There are a lot of different techniques to perform topic clustering [Vay20]. The first topic clustering approach was introduced by Andrew Ng and Michael Jordan in 2003 [Ble03], which is called the LDA. This model is based on a Bayesian model, which reduces the dimensionality of word embeddings and can group words into clusters and differentiate between different topics. Through this approach, a LDA model trained on a corpus of text could infer new text by the topics it found from the training data. The LDA models allow identifying topics in text and modeling the occurrence over time. It is important to note that the output dimension, the number of found clusters, must be decided beforehand. No process automatically decides the correct number of clusters.
After the invention of the LDA, new approaches were published shortly after. Prominent examples are the dynamic and self-aggregating topic models [Vay20]. Nowadays, the state-of-the-art topic clustering algorithms are based on DL, or more particularly, based on transformer models.

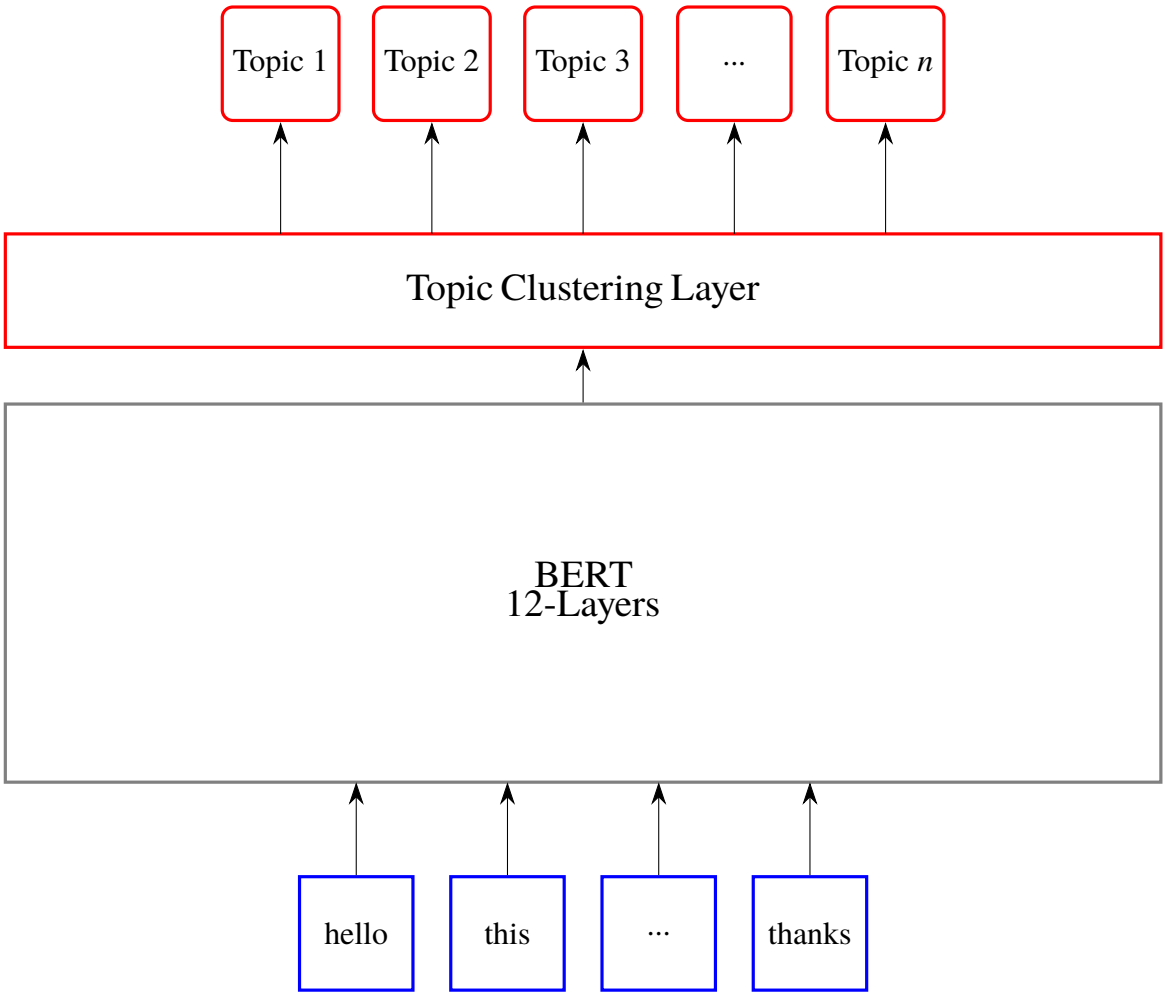


Figure 3.1: High-level view of the BERT [Dev18] topic clustering architecture.

Figure 3.1 illustrates the architecture of the BERT-based topic clustering algorithm. BERT is the core of the entire architecture, whereas the *Topic Clustering Layer* can be trained by fine-tuning, or be replaced with, e.g., a classifier. The rectangles at the bottom represent the input words. Each word of one training sample will be converted into an embedding vector by the BERT 12-Layers. These vectors serve as input for the topic clustering layer, which performs the dimensionality reduction and creates the topics for the predefined number of desired topics, visualized as red rectangles with rounded corners in Figure 3.1. As can be seen, the BERT core is replaceable with any arbitrarily trained BERT model. There exist various models on the Hugging Face website [Huga], which were trained for many different purposes:

BERT Model	**Trained on**	**Application fields**
bert-base-uncased [Dev18]	Wikipedia data	Text generation
RoBERTa [Liu19]	Wikipedia data, books	Feature extraction, topic clustering
bert-base-chinese [pat20]	Books, crawled data	Chinese feature extraction
XLM-RoBERTa [Con19]	2.5TB multilingual data	Multilingual feature extraction
SciBERT [Bel]	1.14M scientific papers	Scientific feature extraction

Table 3.1: BERT [Dev18] models for different application fields taken from Hugging Face.

According to Table 3.1, there are pre-trained BERT models that have been trained in various languages. In addition, the upper layer (*Topic Clustering Layer*) is also replaceable. The basis BERT architecture can be used on its own for custom fine-tuning or the already fine-tuned model for the desired purpose can be used. For the case of topic clustering, already multiple pre-trained solutions are provided for free on the Hugging Face website. It is important to note, that after the fine-tuning was performed, the BERT 12-Layer core is not replaceable anymore. If the transformer was fine-tuned with a BERT model trained only on the English language, other fine-tuning needs to be performed if additional languages or multilingual models are required to perform, for example, topic clustering.

3.4.2 Metrics

The most common metrics to evaluate the performance of the topic clustering algorithms are the *Perplexity*, the *Coherence* [Amo] and the *Normalized Pointwise Mutual Information (NPMI)* [Rüd22]. Greene and colleagues[Gre14] researched to evaluate the stability of

the ideal amount of clusters. Contrary to that, Chang and colleagues[Jon] found out, that the human perception of the ideal amount of clusters outperformed classical metrics, like the *Coherence* [Röd15]. It can be seen, that there is much research on how to select the ideal amount of clusters, but also human intuition can compete with classical metrics. The *Coherence* score is nevertheless an important metric, because it describes how similar the topics are to each other regarding their assigned words. Suppose the number of topic clusters was arbitrarily high. In that case, many similar words could be found in different clusters, because there is not enough training data to find more distinct topics [Amo]. Metrics can implicitly deduce a suitable number of clusters for which all topics have few overlapping words. However, there is no guarantee that those are the ideal number of clusters. For this reason, a combined approach like from Cai and his colleagues [Cai19] was used, consisting of a mix of metrics and human intuition.

3.5 Sentiment Analysis

A sentiment analysis classifies a given text as having either a positive or negative sentiment. The traditional way to accomplish this is by counting the occurring words inside the text and mapping them with a pre-defined sentiment list to be either positive or negative. After counting if there are more negative or positive connotated words in the text, a result score between zero and one, where one is positive and zero is negative, is computed.
Novel sentiment analysis models are, as the topic clustering approach, based on transformer models like BERT [Dev18]. Transformer-based sentiment analysis models omit the traditional word counting, but rather transform the words into embedding vectors and identify the sentiment on embedding calculations, because they were fine-tuned and trained on labels with the BERT architecture.
The Hugging Face platform [Huga] provides pre-trained BERT models that replace the *Topic Clustering Layer* with a fine-tuned layer that performs sentiment analysis. The architecture is similar to Figure 3.1, but with the *Sentiment Analysis Layer* instead of the *Topic Clustering Layer*.

Chapter 4

Methodology

4.1 Data Collection

4.1.1 Podcast Crawling

Do and his team [Phi21] worked out a list of 30 pre-selected healthcare podcasts with a special focus on guest speakers with a relevant background, such as entrepreneurs, innovators or scientists. The pipeline for downloading the episodes was built with the programming language Python. The pipeline inputs one or more RSS feed links. An important feature of RSS is that it also provides the metadata of each episode, such as the publishing date. The publishing date metadata is necessary to model the topics over time. The pipeline comes in two modes: *crawl* and *convert*. The first mode is for downloading all episodes stored in every RSS link sub-sequentially:

```
{
  "Podcast 1": "https://podcast1.com/channels/podcast1.rss",
  "Podcast 2": "https://feeds.someurl.com/feeds/podcast2.xml",
   ...
  "Podcast 30": "https://podcast30.com/channels/podcast1.rss"
}
```

Listing 4.1: Example content of the *rss.json*

The second mode *convert*, converts all non-MP3 file formats into MP3, because the speech-to-text APIs require an MP3 file format. One of the 30 podcasts, namely *The Big Unlock*, did not allow to crawl the publishing dates correctly, so the approximately 100 episodes of this podcast needed to be omitted. If the *XML* file of the RSS-Feed will be updated in the future to crawl the publishing date, this podcast can also be included in the data set.
Since this thesis aims to detect trends of AI in podcasts, only podcasts created from 2015 to 2021 inclusively were considered to be able to make a comparison for the year 2022. All podcast episodes from 2021 needed to be downloaded first, but Do and his colleagues provided all prior episodes.
Each podcast was given an abbreviation of its name to increase its readability. All of the used podcasts with their abbreviation, episode count and starting date are listed in Table 4.1 and will be used in Chapter 5. From 2015 to 2021 inclusively, there are 3,449 episodes as MP3 files. This amount of episodes corresponds to a total size of 1.23 Terabytes with an average of 360 Megabytes per episode and a total length of 60,000 hours of speech for all episodes combined. Each podcast had its folder named after its podcast name, which contained all of the downloaded episodes with the following naming convention:

Episode Number_Episode Title.mp3

for every MP3 file. After downloading all podcast episodes successfully, the function *crawl_metadata.py* was called to parse the metadata from all episodes of all podcasts and store them in a central file called *metadata.csv*. Depending on the download capacity, this entire process can take several hours.

4.1.2 Speech-to-Text API Evaluation

Before performing the speech-to-text transcription, Do and colleagues [Phi21] evaluated the performance of APIs from seven different providers. The best performing API at the time of 2021 was *Microsoft's Azure Speechrecognition* [Mic]. Nevertheless, one limitation of this thesis is the mandatory usage of an open-source solution. For that reason, only Mozilla's Vosk [Alp] and Alphacepheis' DeepSpeech [Moz] will be considered for further evaluation. The first evaluation will be performed in the same manner as Do with the same used versions at that time (October 2020) of DeepSpeech and Vosk, but additionally for the latest versions

Podcast	Abbreviation	Episodes	Starting Year
15 Minutes With The Doctor	1MWTD	41	2017
Beckers Healthcare Podcast	BHP	168	2021
Conversations on Health Care	CoHC	68	2020
Data Book	DB	67	2018
Digital Health Today	DHT	108	2016
Faces of Digital Health	FoDH	51	2017
GeekWire Health Tech	GHT	37	2017
Harlow on Healthcare	HoH	93	2017
Health Care Rounds	HCR	140	2018
HealthChangers	H	64	2017
Healthcare IT Today	HIT	58	2019
Healthcare Rap	HR	79	2019
Healthcare Strategies	HS	92	2019
Healthcare Triage Podcast	HTP	71	2015
Healthcare Weekly	HW-AtFoHI	21	2019
Hospitals In Focus	HIF	54	2018
How I Transformed This	HITT	31	2019
Medtech Talk	MT	141	2015
Outcomes Rocket	OR	803	2017
Perspectives on Health and Tech	PoHaT	121	2018
PopHealth Week	PW	190	2015
Solving Healthcare	SHwDKK	163	2019
Talking HealthTech	TH	179	2018
The #HCBiz Show!	T#S	73	2017
The Change Healthcare Podcast	TCHP	114	2018
The Incrementalist	TI	152	2018
The Pulse by Wharton Digital Health	TPbWDH	69	2019
The Tate Chronicles	TTC	83	2018
This Just In	TJI	118	2015
	Sum	**3,449**	**2015-2021**

Table 4.1: Overview of 29 healthcare podcasts used as data sources for this thesis.

of both APIs as of February 2022 as well. The second evaluation utilizes a more sophisticated approach to enhance the robustness of the evaluation by adding randomness to the calculation.

A commonly used metric to evaluate the performance of speech-to-text API transcriptions is the WER metric [Par08; Err18]. The WER is defined by Zechner [Kla00] as:

$$\text{WER} = \frac{S + D + I}{N} = \frac{S + D + I}{H + S + D} \tag{4.1}$$

where

- S is the total number of substitutions,
- I is the total number of insertions,
- D is the total number of deletions,
- H is the total number of correct words,
- N is the total number of input words,

where each of these five symbols indicates an operation on the text. For example, substitution is the process of removing a word from the predicted transcription, because it was not present in the original transcription. Insertion is the process of adding a word into the predicted transcription because it was missing, but present in the original transcription. The WER is therefore the summation of all operations that have to be performed on the predicted text in order that the text is similar to the original transcription.

To be able to compute the WER, only the audio file itself and no metadata is required. For this reason, all 30 podcasts were used to compute the average WER. For the first WER evaluation, the version used for Vosk was *0.3.15* and for DeepSpeech *0.8.2*. Also the latest versions for Vosk *0.3.32* and for DeepSpeech *0.9.3* as of February 2022 were used for the first approach. The methodology was to first pick one random episode from each podcast. Then, for each episode, three 30-second-snippets, one from the beginning, the middle and the end of each episode were picked. All snippets have been transcribed by hand to be used as the ground-truth label. The WER metric was calculated based on the transcriptions and the ground-truth labels. In the first evaluation, the performance for both Vosk and DeepSpeech for the older versions and new versions shall be measured with the procedure of Do and his team [Phi21]. Other episodes have been randomly chosen compared to the episodes Do and colleagues used. For that reason, new transcriptions by hand needed to be conducted. Figure 4.1 illustrates the first evaluation with the performance of Vosk and DeepSpeech for both the old and new versions, as well as the ground-truth WER computed by Do and his team in October 2020.

Podcast hosts generally start their recordings with the same introductory and farewell phrase,

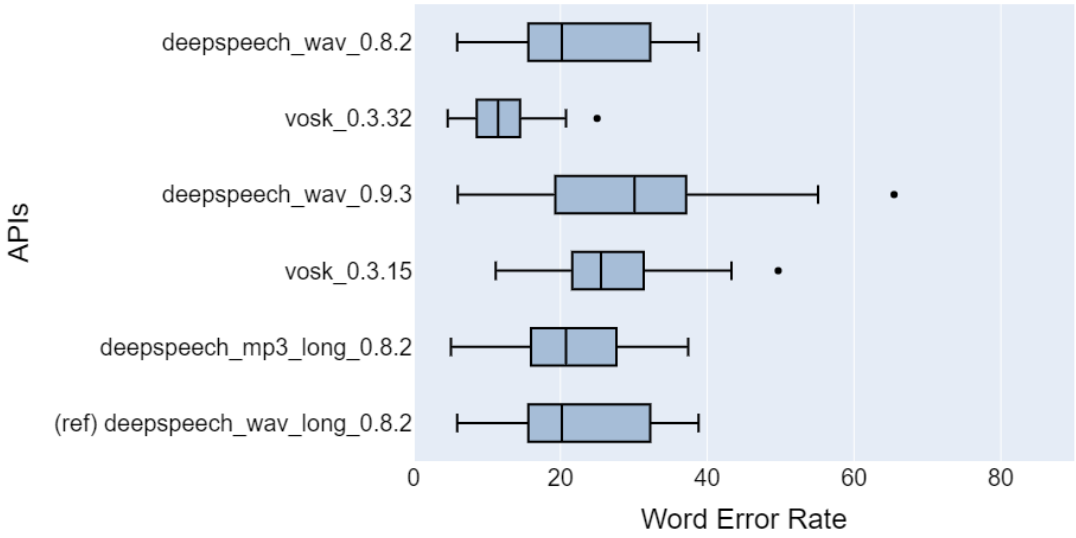


Figure 4.1: WER evaluation according to Do et al. [Phi21] for Vosk and DeepSpeech for the versions from October 2020 and February 2022.

potentially even with music played in the background. To avoid deterministic correlations, a second approach was performed. One random episode for each of the 30 podcasts was also chosen for the adapted methodology. However, the three 30-second-snippets were chosen at a random position for each episode. A constrained guarantees that the three 30-seconds-snippets can not overlap by accident. Each snippet is guaranteed to have a unique word sequence. As in the first approach, the randomly selected 30-second-snippets were transcribed by hand to be used as a label. The three snippets for each episode were first averaged for each episode and then averaged again for each API version. Furthermore, this approach was performed only on the latest versions from Vosk *0.3.32* and DeepSpeech *0.9.3* (as of February 2022) and the performance will not be compared with older versions of both APIs. The version comparison in the first evaluation serves only for the recreation and validation purpose of the research from Do and his colleagues [Phi21]. Nevertheless, the API of choice for Do was DeepSpeech.

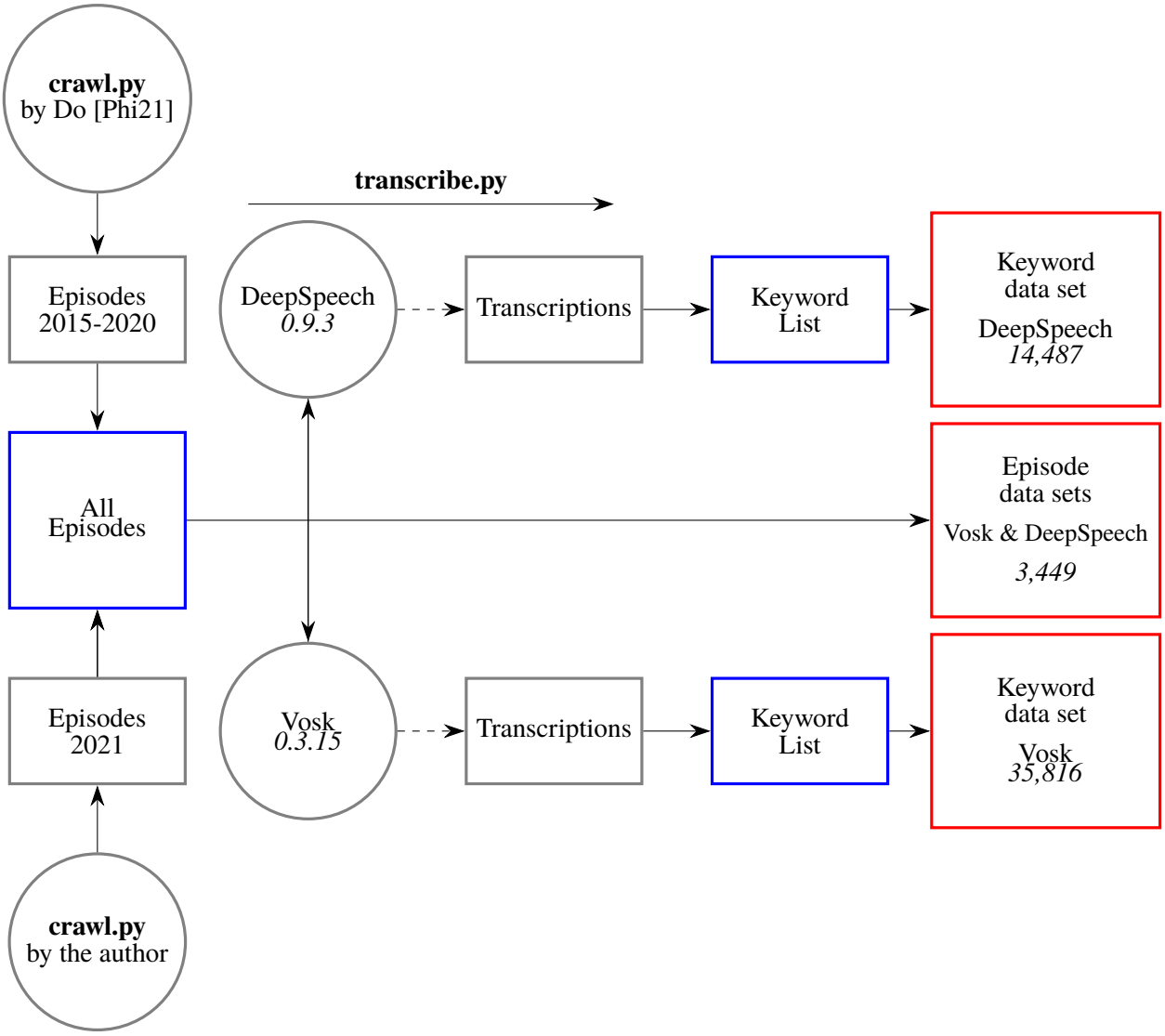


Figure 4.2: Data set generation pipeline with audio crawling, transcriptions and keyword application for all four proposed data sets.

4.1.3 Podcast Speech-to-Text Transcription

The next step is the transcription process for all episodes. Figure 4.2 illustrates the entire pipeline starting from the crawling procedure from Section 4.1.1 until the finalized four training data sets. The pipeline from Do and colleagues [Phi21] includes a **transcriber.py** file, which can be executed in **run** mode. Values need to be added into the constant *SPEECH_API*, which contains the names for the evaluation APIs Vosk and DeepSpeech. The constant *DEEPSPEECH_MODEL* contains the path for the language models and the scorers. These paths can be changed to switch between different versions for Vosk and DeepSpeech, which were used for the WER evaluations. Both APIs require that the language model and the scorer model are downloaded [Moz; Alp] and stored locally. The Python script **transcriber.py** will loop over every single episode, transcribes it for the provided APIs and stores it as **.txt** file into the *podcast* folder, with a sub-folder for each podcast. The 3,449 episodes in Table 4.1 have an average length of approximately 30 minutes. For the transcription of all episodes with Vosk and DeepSpeech, an NVIDIA RTX 3080 Ti with 10,240 Compute Unified Device Architecture (CUDA) [NVIa] cores was used. The more CUDA cores in a GPU, the faster the model training can be finished. Performing the transcriptions on a Central Processing Unit (CPU) would take substantially longer. In the beginning, a problem occurred by trying to use the Graphics Processing Units (GPU) for transcribing with the DeepSpeech API. DeepSpeech requires a specific version for the CUDA driver to be installed on the operating system. The CUDA driver is necessary so the system can operate with the CUDA cores from the GPU. DeepSpeech requires that exactly the CUDA version *10.1* is installed. On a Windows 11 machine, this can be accomplished, for example, by using the *Windows Subsystem for Linux* with *Ubuntu 18.04*. A specified version of CUDA can be installed on this virtual machine without performing a downgrade on the main operating system. Vosk allows to facilitate the GPU for the transcription process more easily than DeepSpeech and no specific CUDA version is required. For that reason, Vosk was chosen as the default API, because more time was needed to identify how the GPU acceleration for DeepSpeech can be acquired. The default results from Chapter 5 are therefore based on the Vosk transcriptions.

The computation time of speech-to-text transcriptions varies a lot by the device that has been used. If an RTX 3080 Ti GPU is used for accelerating the speech-to-text transcriptions, it computes the transcriptions much faster compared to the Ryzen 9 5900X (Table 4.2). The performance is still good for the Ryzen 9 5900X CPU, but considering the thousands of

Device	Performance	Transcription/1h
NVIDIA GeForce RTX 3080 Ti [NVIb]	10,240 CUDA cores	approx. 5 minutes
NVIDIA Tesla K80 [NVIc]	4992 CUDA cores	approx. 15 minutes
AMD Ryzen 9 5900X [AMD]	3.70Ghz x 12 CPU cores	approx. 20 minutes

Table 4.2: Comparison of the speed for different devices for the speech-to-text transcriptions.

hours of audio, the transcription process will take four times more days to complete. Another possibility is Google Colab [Gooa], which provides the free usage of the NVIDIA Tesla K80 from Table 4.2. The downside of this approach is the necessity to outsource the audio files into the Google Cloud [Gooc], which takes time to upload the 1.23 Terabyte of audio files again. Furthermore, the Google Colab computation has an expiration time of 12 hours. This means constant observation, downloading, and uploading of the data is required when using Google Colab. Therefore, if available, the usage of a local GPU is recommended for this process. The entire transcription process with the NVIDIA GeForce RTX 3080 Ti on the 1.23 Terabytes of audio data for the 3,449 episodes with the latest version for Vosk *0.3.32* took approximately two weeks of total transcription time.

4.1.4 Keyword List

The keyword list, sometimes called buzzword list, was also instantiated by Do and colleagues [Phi21] to detect AI-related keywords in the podcast transcriptions. The list was generated by aggregating and filtering many different AI, ML and Data Science glossaries from well-known companies through a pipeline written in Python, called **keyword.py**. The following sources were considered to create the keyword list:

- Accenture Applied AI Glossary [Acc],
- Gartner Hype Cycle [Gar],
- Github ML Glossary [Gitb],
- Google ML Glossary [Goob],
- Microsoft ML Glossary [Mic],

- Stanford Glossary [Sta].

The used keyword glossaries have already been referenced by multiple sources [Ron98; Ril18]. The original buzzword list from Do et al. [Phi21] was refurbished so synonyms and acronyms were eliminated. For example, both the terms AI and Artificial Intelligence were included in the original list, but the acronym AI was eliminated. The finished list contains 102 compared to the original 150 words, which can be found in the Appendix B.

4.2 Data set Building

This section describes all the processing steps on the raw transcribed text before it was used as input to the topic clustering and sentiment analysis models. Most publications about topic clustering perform model training on texts with a short total word length. An increase in corpus length for each training sample has adverse effects on finding the optimal number of topics [Sba20]. Much research was conducted to identify trends from social media texts with topic clustering [Agh18; M21].
Due to the author's knowledge, no research has been conducted yet on topic clustering on continuously transcribed podcast texts. Fast and Horvitz [Fas17] performed topic clustering on the New York Times articles from 1986 to 2016. They divided each article into paragraphs to let humans annotate them. Written articles have a clear structure with ideally one thought per paragraph. Contrary to that, podcast transcriptions are continuous text without a clear structure.
Complying with previous research, using the full episode transcriptions as a training sample has proven not ideal for the related work because each episode has an approximate length of 5000 words. A different strategy was chosen to divide the transcriptions into smaller chunks. This strategy will be explained in the next Section 4.2.1.

4.2.1 Keyword data set

The keyword list was used to find syntactical small but meaningful chunks. Similar abbreviations were also summarized in the keyword list. Before applying the keyword list to the text, a mapping for the words in the raw transcription text was performed. The mapping was performed to comply with the keywords from the keyword list. Table 4.3 shows the abbreviations that have been mapped into the according to keywords.

Abbreviation	Keyword	Hits
ai	artificial intelligence	3348
data set	data	498
gan	generative adversarial network	7
ml	machine learning	115
vr	virtual reality	498

Table 4.3: Mapping from abbreviations to keywords with the corresponding number of hits in the episode texts.

This mapping ensures that often occurring terms are not counted twice. These five mappings from the abbreviations to the keywords have been explicitly identified and performed accordingly to the number of hits given in Table 4.3. At this point, the division in chunk size training samples can be conducted based on the keyword list and the mapped raw text.

4.2.2 Text Cleaning

Topic clustering and sentiment analysis use the embeddings of words as input for their training and inference process. Words that are similar in meaning but differ slightly from each other, e.g., through the upper or lower case, past or present tense or simply by singular and plural, need to be uniformly processed so similar words are not counted twice. A text cleaning pipeline is a crucial step before applying topic clustering or sentiment analysis [Sch19]. The pipeline used for this thesis is stored as the **data set_preparation.py** file in the **scripts** folder. For every episode transcription, the cleaning is performed in the following steps shown in Figure 4.3, where each episode transcription equals the *Raw Text*.

Each of the 3,449 episodes was scanned iteratively for the 102 keywords in the keyword list. A mean occurrence of 4,91 $\pm$ 8,94 keywords for the Vosk transcription was found in every episode. From the raw text, the idea was to use each keyword occurrence and pick a fixed number of words post and prior to it, which will be called text chunk throughout this thesis. The number of words used to create the text chunks was selected by comparing the development of two computed metrics. First, the relative proportion of overlapping text chunks reveals how significant the percentage of overlapping text chunks is within one episode. The closer the number gets to 1, the worse the selected number k for the number of words

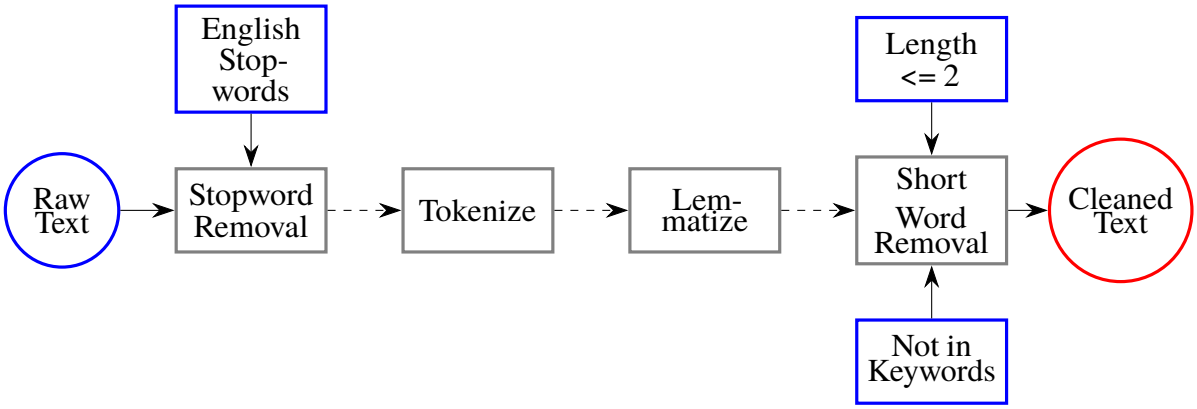


Figure 4.3: Text cleaning pipeline for all four data sets.

prior and post to the keywords gets. The second metric is the relative proportion of used words. This metric tracks how many words of the total number of words per episode are located in all text chunks for each episode. This metric gets steadily closer to one when k it is increasing because it approaches the length of the entire episode text. For this reason, the desired number of k shall be as small as possible, but still preserves as many words from the text. Figure 4.4 illustrates the distribution for both of the explained metrics plotted with the standard deviation as the 95 percent quantile for both metrics.

The mean length of the Vosk episode transcription is approximately 5,000 words, from which on average nine keywords are found. Considering that the two metrics intersect at k equals 150, but on average ten keywords are found within each episode, a size for k of 200 was selected. This size generates a total chunk length of 401, because 200 words prior and post to the keyword are selected. An average preservation rate of approximately 40% is reached with this chunk size. Furthermore, 40% of the text chunks overlap on average.

The scanning of each keyword in the episode text was first performed on the Vosk transcription and later for the DeepSpeech transcriptions. This approach leads to two distinct keyword data sets for Vosk and DeepSpeech. The files were exported into a *.csv* file and used with the Pandas [Pan] package for Python.

The *textfile_name* is built up from the *abbreviation* of the podcast itself and the *episode number*. The cleaned keyword data set for Vosk has a total length of 35,816 entries, whereas the keyword data set for DeepSpeech only has 14,487 data points. The possible reasons behind this substantial difference will be further discussed in Section 6. The Vosk keyword

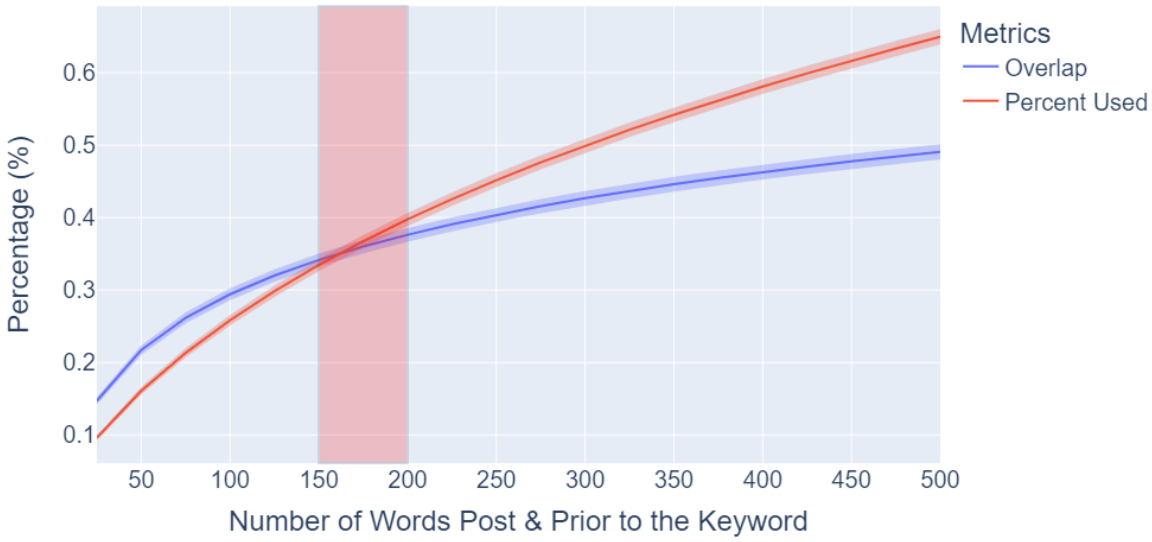


Figure 4.4: Distribution of the relative proportion of used words and the proportion of overlapping text chunks from the Vosk episode transcriptions.

Podcast	**Published**	**Textfile**	**Keyword**	**Raw_Text**
This Just In	2016-02-18	TJI_4	data	in partnership with...
This Just In	2016-02-18	TJI_4	api	been working really...
...	...	...	...	...
Beckers Healthcare Podcast	2021-12-31	BHP_315	data	medication or the...

Table 4.4: Three example rows of the keyword data set from the Vosk transcriptions.

data set was used as the base data set for most of the results and the plots in Section 5.

4.3 Topic Clustering

4.3.1 Training

The BERT architecture from Section 3.3.1 was used for the model training. The pipeline for topic clustering was built with *Optimizing and Comparing Topic Models is Simple (OCTIS)* [Ter21], an open-source technology provided on Github [Gita] which builds upon the Hugging

Face BERT models [Tho20]. The advantage when using OCTIS is, that the metrics *Coherence* and *NPMI* can be calculated off-the-shelf. One drawback is that the inference is not as easily applicable, which will be further explained in Subsection 4.3.4. OCTIS requires three files for the training process:

- the **vocabulary.txt** includes all of the words from the training corpus,
- the **corpus.tsv** contains all the training samples,
- and the **indexes.txt** contains either the label *train* or *test* for each training sample.

These files must be created to use the OCTIS model training. The **corpus.tsv** file has one training sample, in this case 401 words, in every row.
Coming back to Figure 3.1, the BERT model needs to be defined as a parameter before the training can be started. For this purpose, the transformer model *all-mpnet-base-v2* was used, because it provides the best performance for the English language feature extraction [Jay21]. The *Topic Clustering Layer* from Figure 3.1 is provided off-the-shelf from OCTIS. It was already pre-trained and only the BERT model remains interchangeable. Now the **preprocess_data set** function from OCTIS can be called to build the following files:

- **bert_path_train.pkl**,
- **bert_path_test.pkl**,
- **bert_path_val.pkl**.

These three files store the embedding vectors created by the BERT 12-Layer architecture. Depending on the size of the data set, the computation of this embedding transformation can take up to several hours.

4.3.2 Number of Topics Selection

The ideal number of topics is an important parameter because based on this number the decision of how many topics will be used for the training data is made. If the number is too low, the entire context of the training data could not be caught. Topics that are desired to be found, such as AI-related topics, could accidentally be omitted by the model. If the

defined number of topics is too high, a lot of overlapping topics will occur and the topics over time analysis could fail. Therefore, the metrics from OCTIS [Ter21], namely *Coherence* and *NPMI* were used as a basis estimation for the perfect number of topics. The metrics were calculated iteratively from the range [10, 40] with a step size of two, as can be seen as the x-axis in Figure 4.5. The product of both metrics was also calculated to identify the number of topics with the highest overall score. The product was multiplied by ten so that the values are also visible in Figure 4.5. It can be seen, that the perfect number of topics for the Vosk keyword data set is 14. This value for n will be used to optimize the embeddings further.

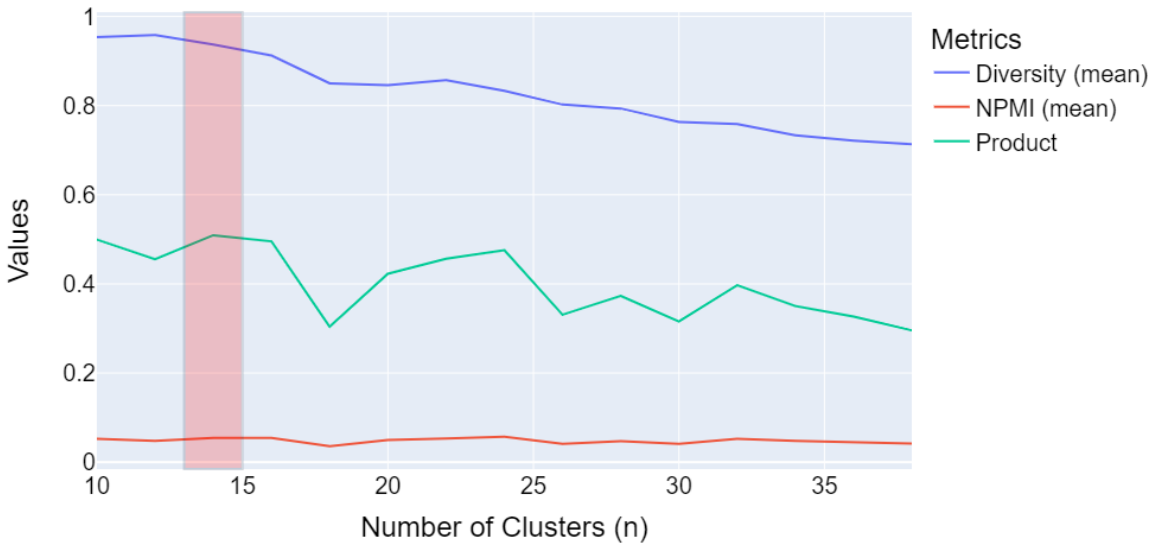


Figure 4.5: The optimal number of topics for the Vosk keyword-level data set.

4.3.3 Optimization

Another aspect why OCTIS [Ter21] was chosen as the default technology for performing topic clustering, is the easy usage of the hyper-parameter optimization function. A metric and a search space can be defined to perform Bayesian optimization [Sha15]. The optimization result provides:

- best-seen values of the chosen metrics with the corresponding hyper-parameter configurations,

- hyper-parameters,
- metric values,

for each optimization iteration itself. The optimization can be performed by calling the *Optimizer.optimize* function with the desired metrics and search space to optimize on. The metrics used for the topic clustering were the default metrics provided by OCTIS, namely the *Coherence* and the *F1-Score*. The optimization can only be applied if a test data set is provided for the evaluation of the *F1-Score* [Faw06]. For this reason, a random subset of 20% was chosen from the data set to be used as test set for the optimization. The search space used for optimization is the default search space provided by OCTIS [Ter21]:

- *num_layers*: Categorical(1, 2, 3),
- *num_neurons*: Categorical(100, 200, 300),
- *activation*: Categorical('sigmoid', 'relu', 'softplus'),
- *dropout*: Real(0.0, 0.95).

Furthermore, the number of *model_runs*, *optimization_runs* and *number of topics* needs to be set. Due to the great computing power necessary to perform this task, the *model_runs*, which is responsible for averaging multiple runs, was set to two. The *optimization_runs* parameter was set to 50, which can be viewed as the number of epochs trained or optimized in this case. The number of topics was calculated in the last step and accordingly chosen as a parameter for optimization.
Another critical note is that the Vosk keyword data set has a size of 35,816 training samples with 401 words each. It was necessary to have 64 Gigabyte of Random Access Memory (RAM) installed, otherwise the optimization could not be performed. The whole optimization with the NVIDIA RTX 3080 Ti took approximately five hours.

4.3.4 Inference

After training the model with the hyper-parameters from the optimization function and the ideal amount of clusters (14 in this case), the model training returned a large dictionary. The dictionary contains the following entries after the training with the 35,816 by 401 Vosk keyword data set:

- *topics*: (14 x 10) the list of the most significative words foreach topic (list of lists of strings),
- *topic-word-matrix*: (14 x 25,070) matrix of weights where 14 is the number of topics and 25,070 is the vocabulary length,
- *topic-document-matrix*: (14 x 35,816) matrix of weights where 14 is the number of topics and 35,816 is the number of documents in the corpus.

Each topic contains a list of ten words in string format. The naming convention was to use the first five topics and connect them with an underscore together so that a single string could describe each topic. Each of the 14 topics is defined by the ten most contributed words to this topic. The 14 topics have words connected to them, but no *generic term*. The generic term needs to be self-defined by hand for each topic. In this case, the assigned generic terms for the 14 topics can be seen in Table 4.5. The *topic-document-matrix* is essential to identify the representing topic for each training sample. For each sample, OCTIS [Ter21] computed one vector of the length of the number of desired topics, here 14. Each of the 14 scalar values represents how present each topic is in the current training sample, accumulated to one.

Topic String	**Generic Term (Topic)**
system_record_information_provider_need	electronic health record
cloud_think_know_market_vendor	cloud architecture
software_company_start_bit_work	startup acceleration
triplet_trunk_nexus_invisibility_bidding	unknown
privacy_security_share_hipaa_blockchain	data privacy
datum_data set_source_data_research	etl pipeline
care_patient_population_community_hospital	home care
machine_algorithm_learn_ai_image	**ai**
pressure_blood_phone_day_minute	heart monitoring
podcast_today_leader_healthcare_join	guest speakers
cost_transparency_price_pay_payer	hospital pricing
clinical_trial_drug_disease_cancer	data privacy
vaccine_kid_virus_people_covert	vaccination
technology_healthcare_platform_solution_need	healthcare innovation

Table 4.5: Mapping from the topic string to the self-defined *generic term* for all 14 topics with the Vosk keyword-level transcriptions.

The highest value for each training sample was chosen as the most dominant topic. Additionally, the embeddings from the Vosk keyword data set can be used to perform inference on the episode-level data set. However, in order to use new data that was not included in the training set, the following hack needs to be accomplished to use the inference:

OCTIS provides a function called *inference*, that requires a data set as parameter with a specified OCTIS file format. For the creation of this data set, an extra column with labels is required, even though they are not needed in the case of inference. The labels, either only *train* or *test* must be used, to keep the data together, otherwise the train and test labeled data will be split into different data sets. After building the OCTIS data set, the order of the data remains the same, so the assignment for each predicted topic and episode can be conducted straightforwardly.

The goal is to predict the most dominant topic for each of the 3,449 episodes based on the Vosk keyword data set embedding vectors. After creating the OCTIS data set for the 3,449 episodes, the *hacked* inference was conducted. All labels were defined as *train*, so the topic predictions can be found in the entry *topic-document-matrix* of the resulting dictionary. The topic will be assigned to each episode according to the highest scalar value in each vector. The same procedure was performed for DeepSpeech at a later stage of this thesis.

After performing multiple topic clustering training and inference procedures, four distinct data sets with their topic predictions were generated. This entire process and the resulting data sets were already visualized in Figure 4.2, as well as listed below:

- Vosk keyword data set (35,816): one topic per sample predicted,
- DeepSpeech keyword data set (14,487): one topic per sample predicted,
- Episode-level data set (3,449): one topic per sample predicted with the **Vosk embeddings**,
- Episode-level data set (3,449): one topic per sample predicted with the **DeepSpeech embeddings**.

4.4 Sentiment Analysis

4.4.1 Transformers

The sentiment analysis was performed similarly to the topic clustering, but no on-top technology like OCTIS [Ter21] was used. The Hugging Face platform [Huga] provides multiple different sentiment analysis fine-tuned models. A limitation is that the transformer must be trained in English because the podcast transcriptions are also in English. Therefore, the most popular sentiment analysis transformer for the English language was chosen: *distilbert-base-uncased-finetuned-sst-2-english* [Huga]. A common technique to optimize a DL model is distillation, which reduces the size of a neural network by cutting out uninformative parameters. This model is based on the *distilbert-base-uncased* BERT model (distilled BERT model), which was trained on many random English Wikipedia pages. This model is essentially the same as the *bert-base-uncased* model from Table 3.1, but with fewer parameters to be smaller and more easily deployable. Nevertheless, the performance is almost the same as the full model, because only unnecessary parameters were removed from the model.
The transformer for sentiment analysis can be downloaded and used through the Hugging Face Github package called *transformers* [Hugb]. The module *pipeline* is then used to download the model. The full command to download the model is:

```
from transformers import pipeline

sentiment_classifier = pipeline(
    'sentiment-analysis',
    model='distilbert-base-uncased-finetuned-sst-2-english'
)
```

Listing 4.4.1 illustrates how the transformer for the sentiment analysis was downloaded.

4.4.2 Inference

After downloading the transformer, the usage to make an inference with the sentiment analysis is straightforward:

```
sentiment_classifier ('This thesis is awesome!')
```

Listing 4.4.2 shows, that the *sentiment_classifier* only requires the parameter for the string. This one line of code returns either *POSITIVE* or *NEGATIVE* depending on the sentiment of the inputted string. In this way, the sentiment was calculated for the following data sets:

- Episode level data set (3,449) overall sentiment per sample predicted on the Vosk transcribed text,
- Episode level data set (3,449) overall sentiment per sample predicted on the DeepSpeech transcribed text.

4.4.3 Sentiment over Time

The overall sentiment per podcast and per topic was calculated by dividing the positive example through the total amount of positive and negative predictions. The proportion of positive to negative sentiments was calculated with the following formula:

$$Score = \frac{positive}{positive + negative}, \tag{4.2}$$

where the total number of positive and negative sentiments are divided by their relative occurrence. A score of zero indicates a total negative sentiment, whereas a score of one represents a genuine positive sentiment. This proportion helps to indicate how positive or negative a topic is over a specific period of time.

Chapter 5

Results

5.1 Data Set Overview

The literature research in Section 2 has shown, that podcasts are an emerging and novel research medium. Not only because podcasts differ from classical media like radio, but also because the podcast production has been heavily increasing since 2015. Figure 5.1 illustrates the amount of produced podcast episodes for all 30 podcasts from the year 2015 until the end of 2021. It can be seen, that by the end of 2021 the number of episodes doubled compared to 2020. All 30 initial podcasts are visualized, because no metadata is necessary for plotting the number of episodes. However, all plots that have been generated starting from Section 5.3, only 29 (without *The Big Unlock*) podcasts were used according to Section 4.1.1.

Figure 5.2 shows the total number of produced episodes from 2015 until the end of 2021 for each of the 30 podcasts. It should also be considered, that only four podcasts have started to record episodes starting from 2015, as shown in Table 4.1. This timeframe of podcast production accounts for 3,449 episodes with a total length of approximately 1,700 audio hours. Furthermore, the total word count for all episodes in the Vosk transcriptions is approximately 17 million. The frequency of produced episodes varies significantly by each podcast. It can be seen, that the podcast Outcome Rockets (OR) has the most produced episodes and therefore also accounts for most of the audio hours, approximately 400 hours. The number of episodes for the remaining podcasts is relatively normally distributed.

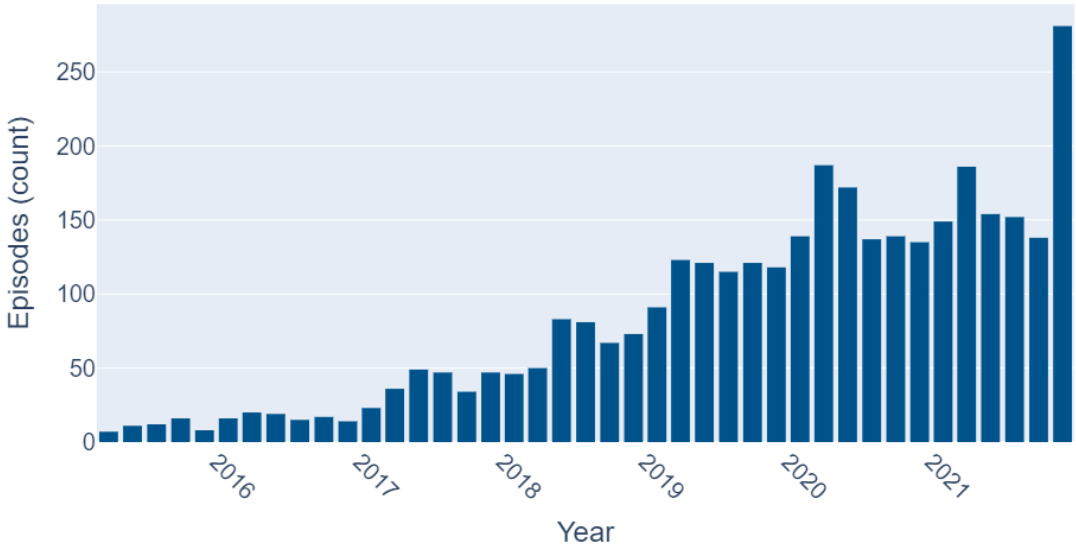


Figure 5.1: Distribution of the number of podcast episodes from 2015 until the end of 2021.

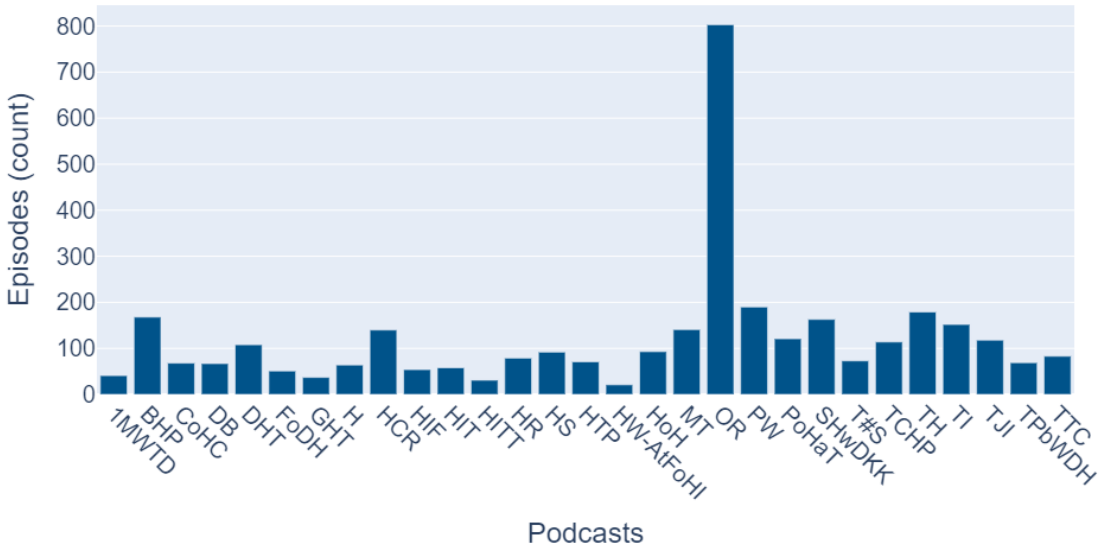


Figure 5.2: Distribution of the number of episodes for each podcast from 2015 to 2021.

5.2 Speech-to-Text API Evaluation

The first research question concerns about the comparison between the quality of the speech-to-text APIs Vosk and DeepSpeech for transcribing podcast audio data. Two approaches were performed to measure the quality of both APIs. Since no metadata is required to compute the WER, all 30 podcasts were used for the WER calculation. The first speech-to-text comparison (Figure 4.1) was the recreation of the approach from Do and colleagues [Phi21] for the latest versions of Vosk and DeepSpeech. The second and more robust approach with additional randomness was only performed on the latest versions of both APIs, because the latest versions have proven to perform on average better according to the comparison in Figure 4.1. For the second approach, the total mean value for each of the three snippets for both APIs, as well as the average over all snippets for each API is illustrated in Figure 5.3.

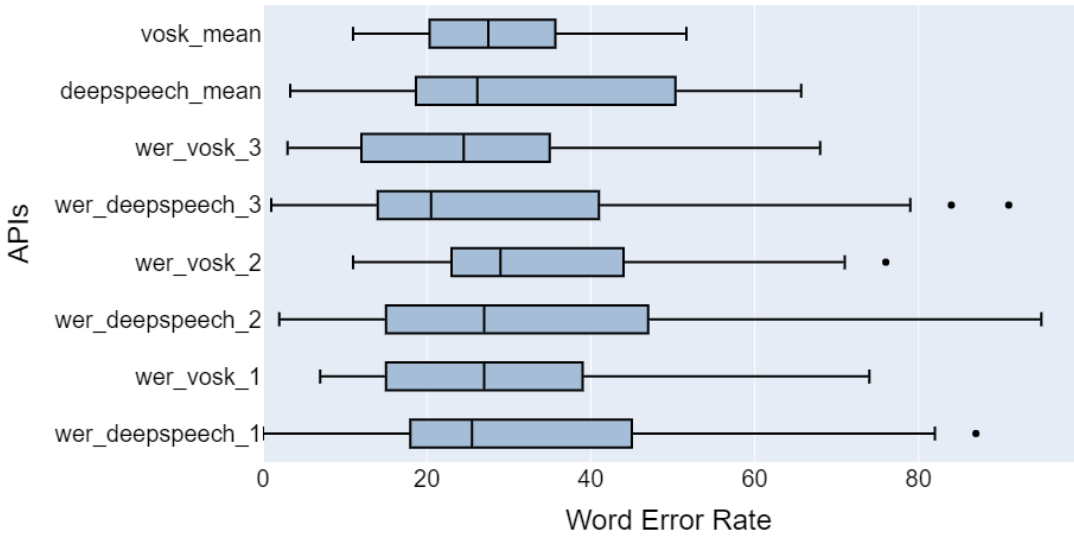


Figure 5.3: Second approach for the WER evaluation with randomness with the overall mean values for both APIs and the mean values for each of the three snippets.

Both mean values over all snippets and episodes for Vosk and DeepSpeech are shown in the top two rows of the boxplot in Figure 5.3, where Vosk has a mean value of 28.95 and DeepSpeech of 29.54. The lower six boxes correspond to each of the three snippets for each of the both APIs. For example, *wer_vosk_3* corresponds to the average WER for the third snippets of each single episode randomly chosen for each of the 30 podcasts transcribed

by Vosk. Contrary to that, the *wer_deepspeech_2* corresponds to the average WER for the second and therefore the middle snippets of each single episode randomly chosen for each of the 30 podcasts transcribed by Vosk. The boxplots show, that DeepSpeech performs on average better than Vosk. Furthermore, the fluctuation of the mean WER values for the second approach in Figure 5.3 compared to the first approach shown in Figure 4.1, is on average less and therefore the second newly proposed approach is more robust.

5.3 Topic Clustering

5.3.1 Co-Occurrence Matrices

The co-occurrences for the keywords were calculated on the episode-level data set transcribed by Vosk with 35,816 text file chunks and the episode-level data set transcribed by DeepSpeech with 14,487 text file chunks, as shown in Figure 4.2. For each of the 3,449 episodes, the keywords from the keyword list (Table B) that occurred in all episodes were counted. Figure 5.4 illustrates which keywords frequently occurred together in one episode. There are 102 keywords in total, but only 23 of these keywords occurred in all of the transcribed texts. The scale is a color bar with a range from 0 to 1, where 0 accounts for no co-occurrence at all and 1 accounts for a high co-occurrence. The more often the keywords appear together in one episode, the closer the value is to 1 and the brighter the block becomes in the heatmap. It can be seen, that *data* is the most common keyword that has been found in the Vosk transcriptions. The keyword *data* often occurs together with all of the other keywords. Also, for the keyword *software* it can be seen, that it often occurs together with keywords such as *api* or *cloud*. Furthermore, *api* is often combined with the keyword *transparency*.
The co-occurrences for the topics were calculated in the same way as for the keywords. For each of the 3,449 episodes, all topics from the *Vosk Embeddings* that occurred together in each episode were counted. Figure 5.5 illustrates the topics on the x and y-axis. The topic co-occurrence diversity is very high, because many of the 14 topics frequently appear together in podcast episodes. The topics *cloud architecture* and *data privacy* frequently occur together with all other topics, which are visualized again by the brighter squares. Another example is, that the topic *vaccination* rarely occurs together with other topics except for *heart monitoring* and *data privacy*. In general, it can be seen that the diversity of co-occurring topics is higher than for co-occurring keywords. The keyword co-occurrence matrix is on average very dark,

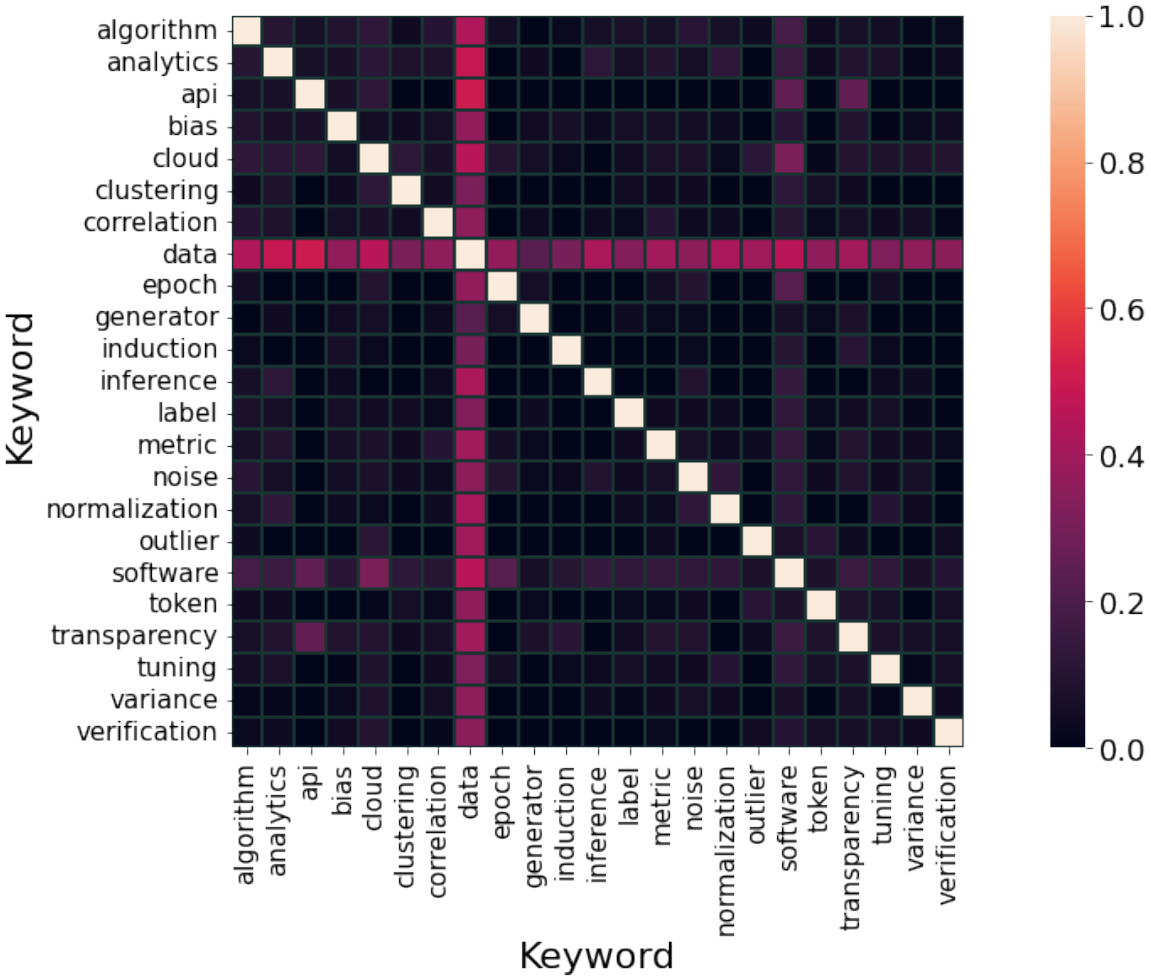


Figure 5.4: Co-occurence matrix for the keywords in the Vosk transcriptions.

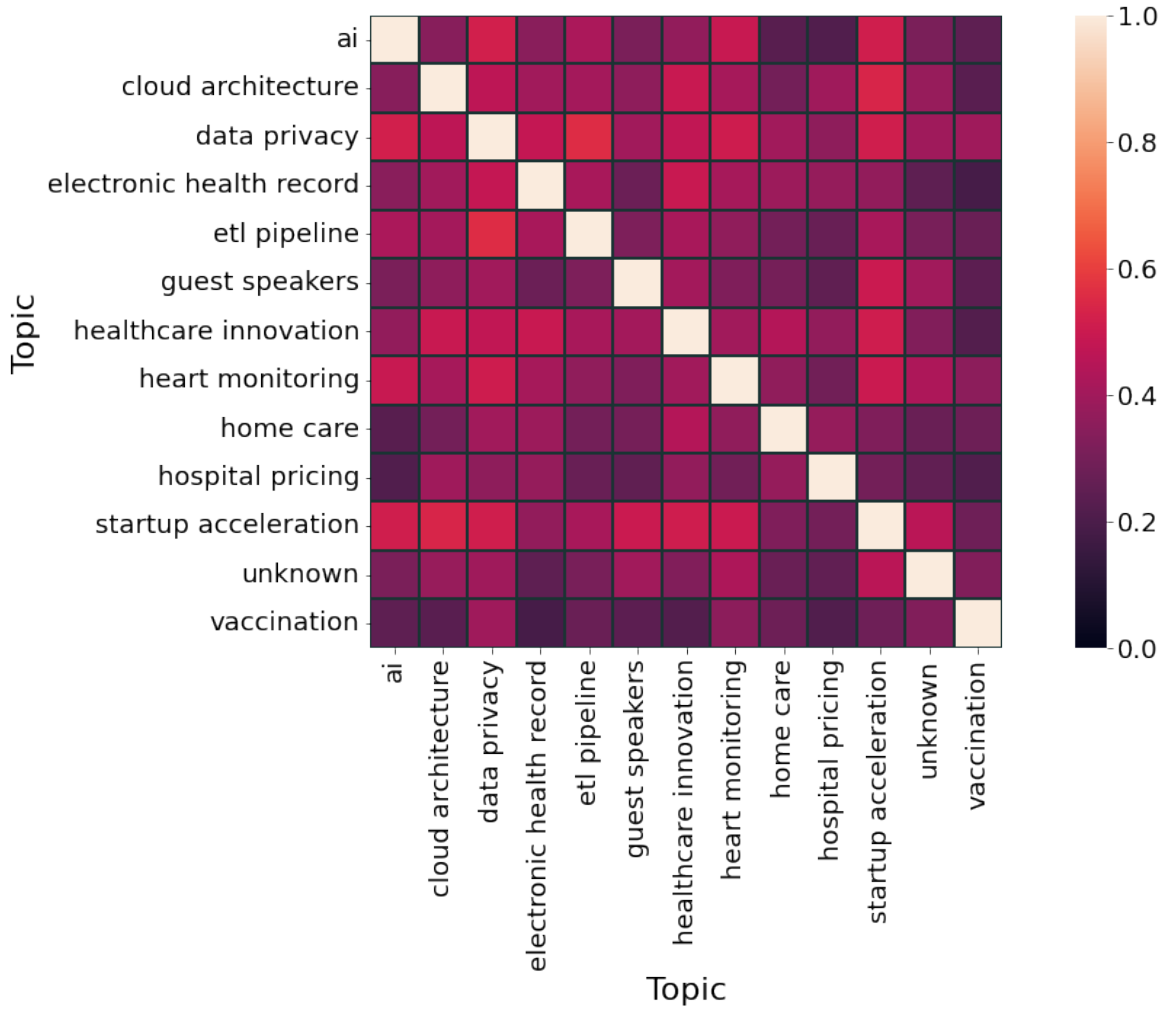


Figure 5.5: Co-occurence matrix for the topics in the Vosk transcriptions.

which indicates that not many keywords are co-occurring together. On the other hand, the topics on the co-occurrence matrix show many different shades of red within the range of the color bar. This indicates, that many topics are co-occurring together.

5.3.2 Cosine Similarities

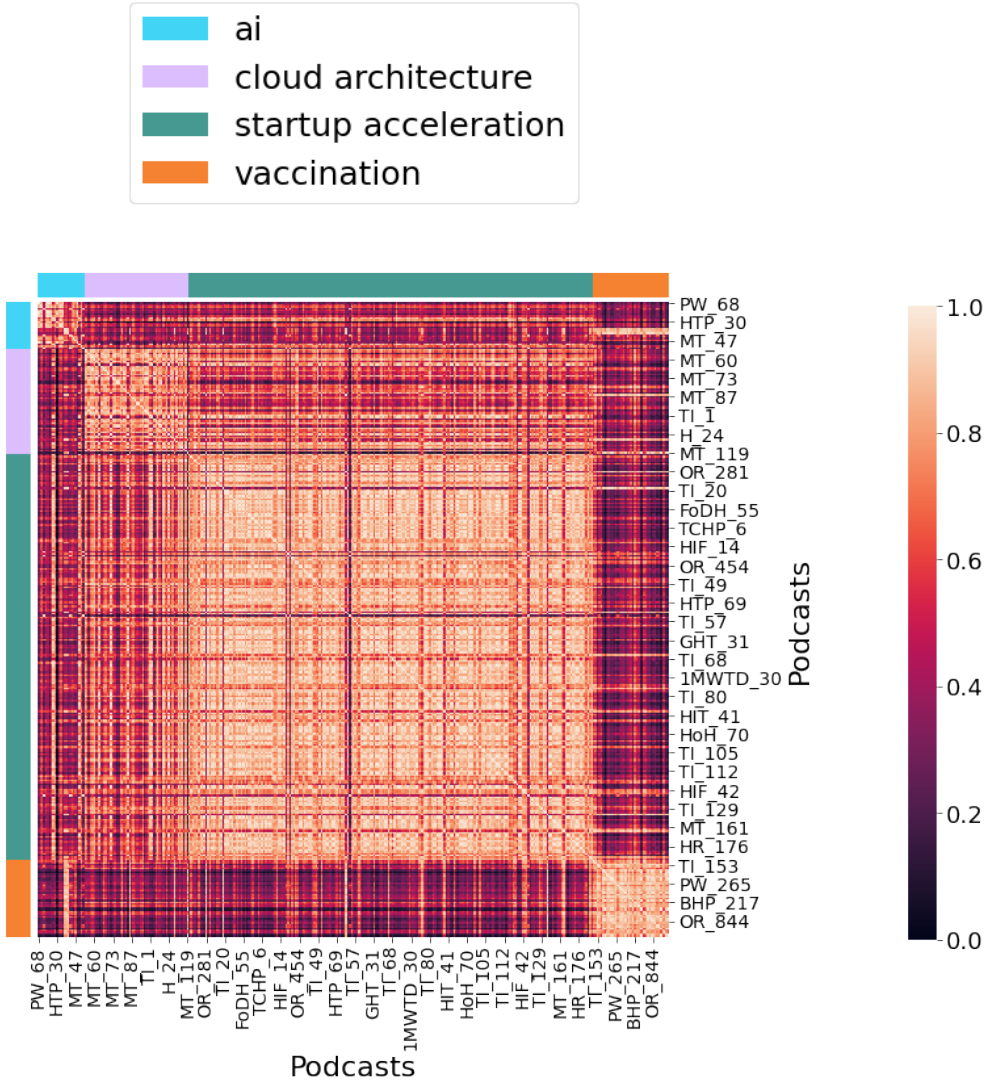


Figure 5.6: Cosine similarity of the Vosk episode-level transcriptions for the topics cloud architecture, ai, startup acceleration and vaccination.

The cosine similarity was calculated to reveal which episodes and podcasts talk about similar topics. If all 14 topics were visualized in one plot, the amount of information could not be visualized descriptive enough. For that reason, a selection of four relevant topics of the current time, namely *cloud architecture*, *ai*, *startup acceleration* and *vaccination* were chosen for the representation purpose. The topic *vaccination* was chosen because of the relevance of the

COVID-19 pandemic at the time in 2022 [Pla20]. The other three topics were chosen, because they are related to the AI-field, which is the topic of this thesis. Figure 5.6 shows these four topics with color bars in the top row and leftmost column. The x and y-axis illustrate a subset of the episodes that contain these four topics. Four distinct clusters have been formed on the heatmap. The biggest cluster is located in the middle of the plot, which represents the topic *startup acceleration*, as can be deduced from the orange color in the top row and leftmost column. The three tinier clusters in the plot's top left and bottom right denote the remaining three topics. Further can be observed, that episodes from different podcasts have a similar cosine similarity towards specific topics. For example, the topic *ai*, visualized with the color green, includes episodes from the podcasts:

- Episode 68 from PopHealth Week (PW),
- Episode 30 from Healthcare Triage Podcast (HTP),
- Episode 47 from Medtech Talk (MT).

These three episodes from different podcasts have been assigned to the same topic *ai*. This example shows the opportunity to find all episodes with similar topics in the data sets. Therefore, this approach can be continued for all topics that have been found in the data sets both on episode-level as well as on podcast-level to find episodes and podcasts that are discussing similar topics.
All embedding vectors from the topic clustering with OCTIS [Ter21] on the Vosk keyword data set were summed up for all episodes of each podcast. The summation revealed the most dominant topics for each podcast and clustered similar podcasts together. The cosine similarity was also calculated at the podcast-level to see which podcasts share the highest similarity.
Figure 5.7 shows, that *healthcare innovation* is the most frequent topic among all podcasts, as visualized as the biggest cluster in the center of the plot. Tinier clusters in the top left and bottom right correspond to the remaining topics. It can be seen, that no podcast has the topic *ai* as the most dominant topic.
A comparison between the topics of the Vosk and DeepSpeech transcriptions on podcast-level can be seen in Figure 5.8. The abbreviations for this topics were also used so that the illustration looks more clear. The four changed abbreviations can be found in Table 5.1.

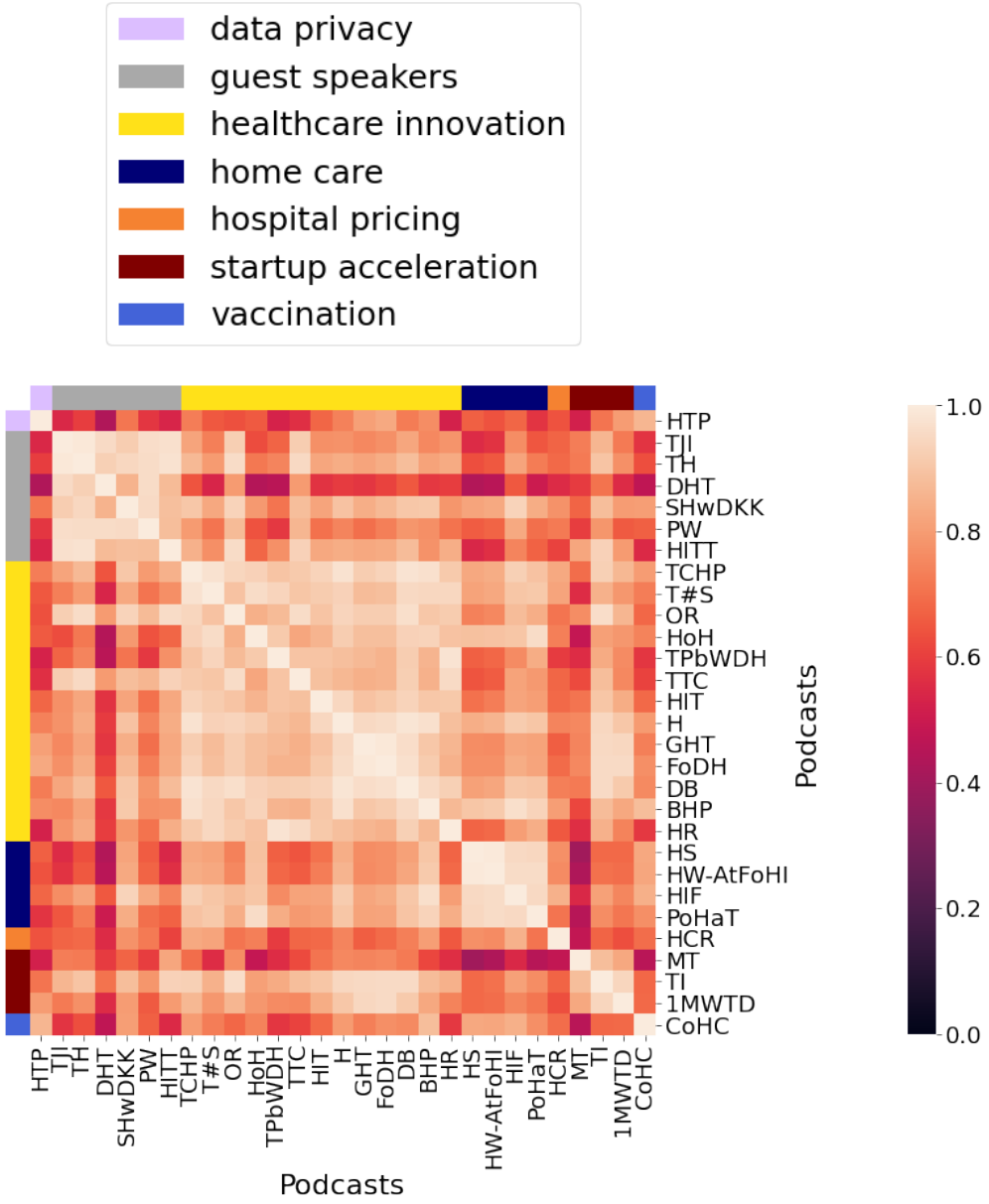


Figure 5.7: Average cosine similarity of the Vosk transcriptions for the most dominant topic of each podcast.

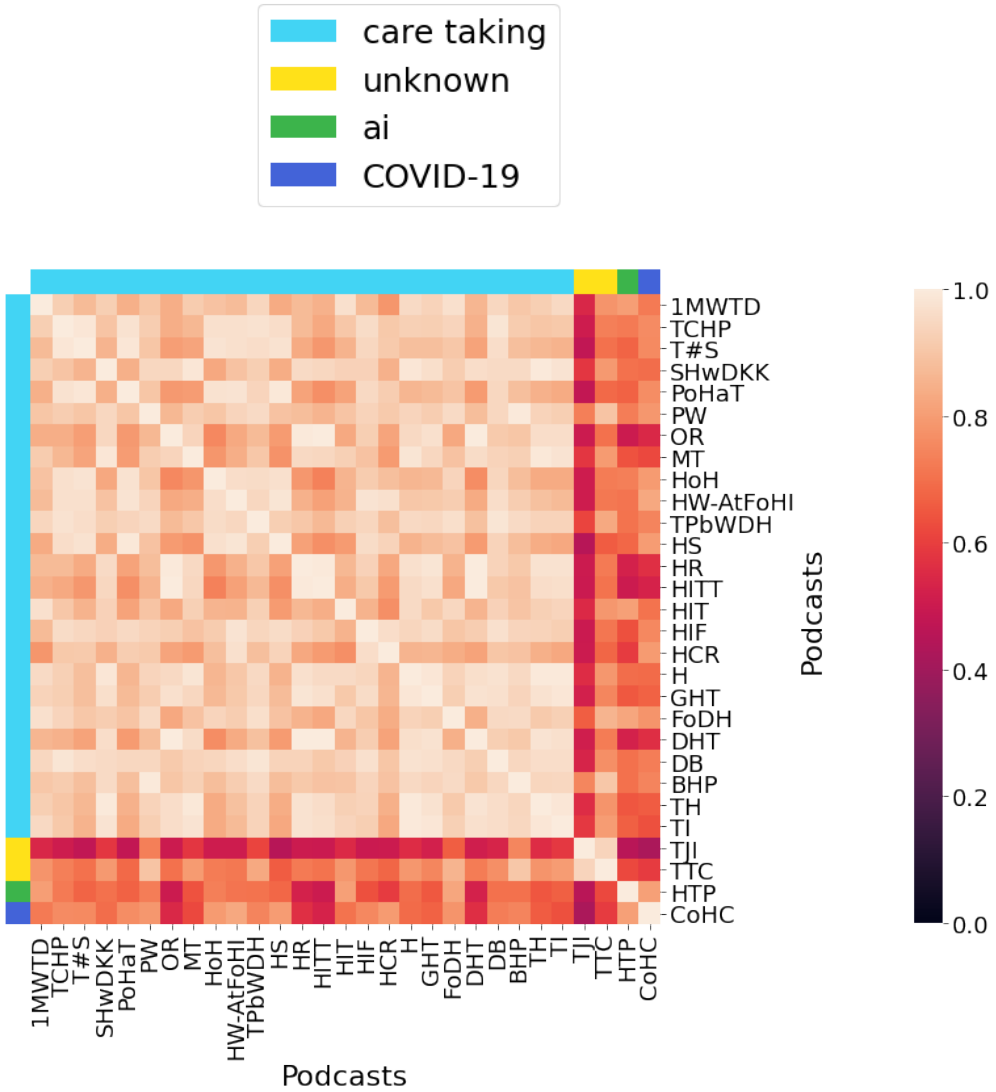


Figure 5.8: Cosine similarity of the DeepSpeech podcast-level transcriptions for the four most dominant topics.

Topic String	**Generic Term (Topic)**
overstone_breeding_dawdle_hunterian_thenglysshman	unknown
vaccine_virus_variant_severe_variance	COVID-19
focus_work_care_driver_leader	care taking
train_training_computer_prediction_doctor	ai

Table 5.1: Mapping from the topic string to the self-defined *generic term* for all 14 topics in the Vosk transcriptions.

This figure illustrates the topics in the DeepSpeech transcriptions for the episode-level transcriptions. The topic *care taking* is the most dominant topic among the four occurring topics. For Figure 5.6, the topic model can distinguish podcasts based on their average topic on episode-level transcription with six distinct topics. Contrary to that, with the DeepSpeech transcriptions, the topic model can only classify four distinct topics for all dominant topics averaged over all episodes. All podcasts with their according topic assignment can be viewed in Table 5.2.

Topic	Podcast	Abbreviation
data privacy	Healthcare Triage Podcast	HTP
guest speakers	This Just In	TJI
	Talking HealthTech	TH
	Digital Health Today	DHT
	Solving Healthcare	SHwDKK
	PopHealth Week	PW
	How I Transformed This	HITT
healthcare innovation	The Change Healthcare Podcast	TCHP
	The #HCBiz Show!	T#S
	Outcomes Rocket	OR
	Harlow on Healthcare	HoH
	The Pulse by Wharton Digital Health	TPbWDH
	The Tate Chronicles	TTC
	Healthcare IT Today	HIT
	HealthChangers	H
	GeekWire Health Tech	GHT
	Faces of Digital Health	FoDH
	Data Book	DB
	Beckers Healthcare Podcast	BHP
	Healthcare Rap	HR
home care	The Change Healthcare Podcast	HS
	Healthcare Weekly	HW-AtFoHI
	Hospitals In Focus	HIF
	Perspectives on Health and Tech	PoHat
hospital pricing	Health Care Rounds	HCR
startup acceleration	Medtech Talk	MT
	The Incrementalist	TI
	15 Minutes With The Doctor	1MWTD
vaccination	Conversations on Health Care	CoHC

Table 5.2: Assignment of each podcast to its abbreviation and to its most dominant topic averaged over all episodes in the Vosk transcriptions.

5.3.3 Topic Development over Time

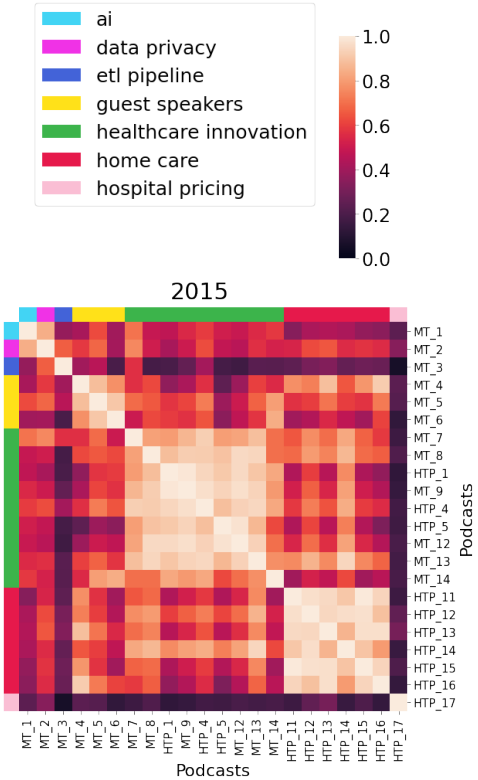
Another analysis was conducted on the topic development over time. Three random podcasts were chosen that started in 2017 or earlier to provide data for most of the years between 2015 and 2021. If a picked podcast has started to record close to 2022, it would have been not applicable to visualize the development over time starting from 2015. Therefore, the picked podcasts are:

- *Medtech Talk*,
- *Healthcare Triage Podcast*,
- *15 Minutes With The Doctor*.

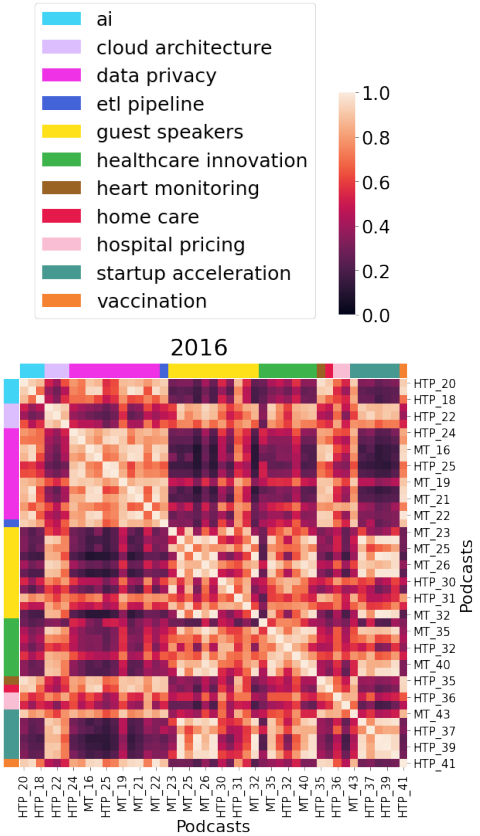
The change over time for the topics of the episode-level transcriptions from these three podcasts are illustrated in the Figures 5.9 and 5.10. The podcasts were grouped into bins from 2015 until 2021. Every year and the average topics over every year were visualized in form of a heatmap.
From 2015 until 2021, the number of topics fluctuated significantly each year. The topic *ai*, as shown in green in Figure 5.10, is always a minor topic, but continuously present from 2015 until 2019. In 2020 no AI-related topic was present. Another example is the topic *healthcare innovation*. This topic faced a hype during 2015 and 2016, but steadily decreased towards 2021. Contrary to that, the topic *data privacy* increased more and more during the course from 2015 until 2021. The average value over every year shows, that the topics:

- *data privacy*,
- *guest speakers*,
- *healthcare innovation*,
- *startup acceleration*,

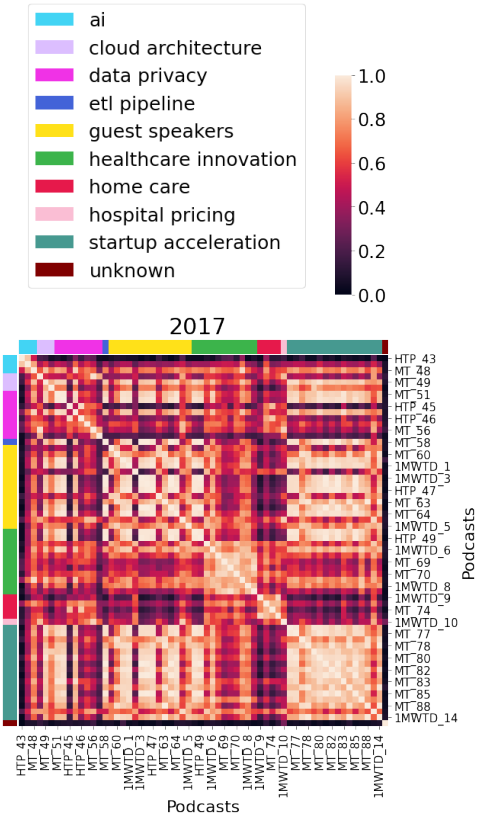
are the most dominant four topics throughout the years 2015 until the end of 2021.



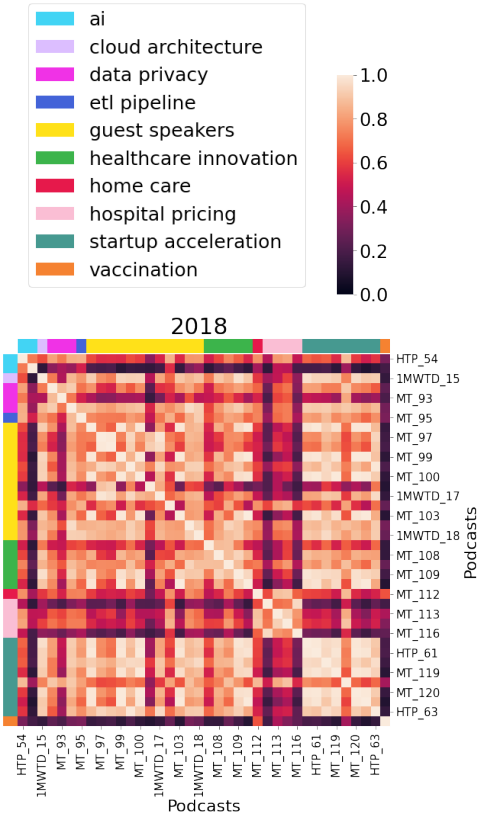
(a) Cosine similarities for the year 2015.



(b) Cosine similarities for the year 2016.

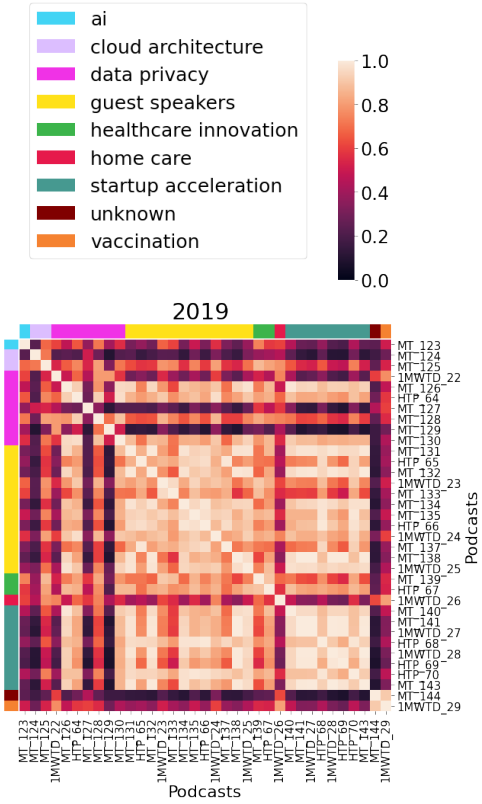


(c) Cosine similarities for the year 2017.

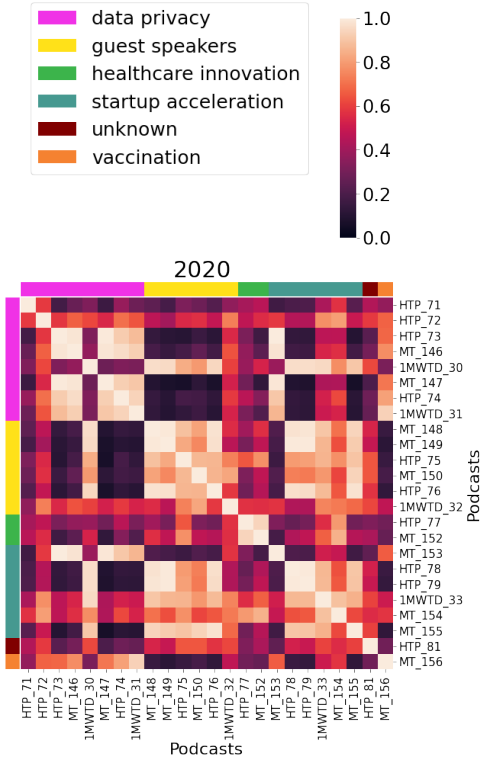


(d) Cosine similarities for the year 2018.

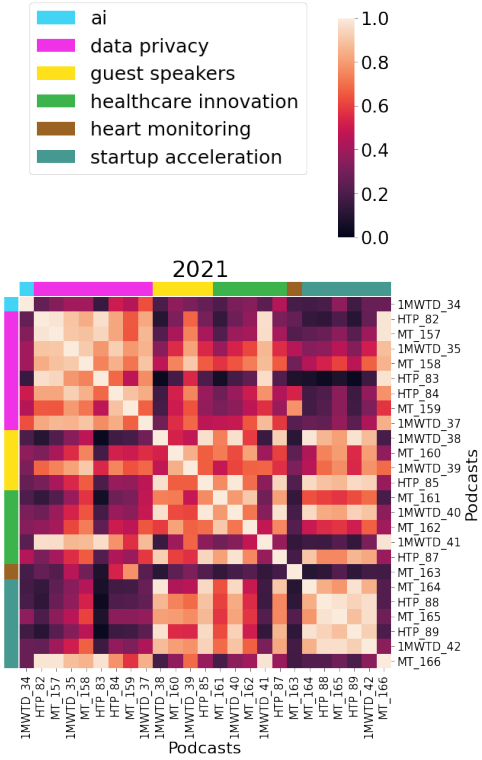
Figure 5.9: Cosine similarities for the topics of the podcasts *MT*, *HTP* and *1MWTD* from 2015 to 2018.



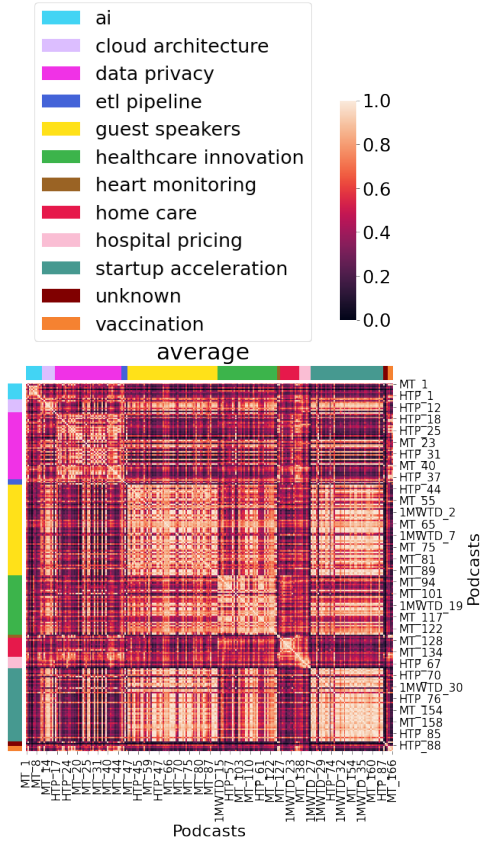
(a) Cosine similarities for the year 2019.



(b) Cosine similarities for the year 2020.



(c) Cosine similarities for the year 2021.



(d) The average cosine similarities over all years.

Figure 5.10: Cosine similarities for the topics of the podcasts *MT*, *HTP* and *1MWTD* from 2019 to 2021 and the overall average.

5.3.4 Subtopic Clustering

An experiment was conducted to find subtopics within the domain of the Vosk text chunks that were assigned to the topic *ai*. From the original 35,816 Vosk text chunks, 2,206 chunks were identified to be assigned to the topic *ai* by the topic model. This chunks have been used to apply the same pipeline as for the episode-level 35,816 Vosk text chunks from Chapter 4. However, the results are highly experimental and shall only serve illustrative purpose and for a potential future experiment. Therefore, some of the found subtopics are discussed in Section 6.3.1 and all of the subtopics can be seen in the Appendix C.

5.3.5 Keyword Search

The transformer model on the keyword keyword-level dataset can also be used to search for novel AI-related keywords in the dataset. Since 14 topics have been identified, from which one topic is AI-related (as can be seen in Table 4.5), this topic can be used to search for novel keywords. The raw episode-level text transcribed by Vosk was used for this purpose, because all of the spoken words are still within the raw episode texts. For each episode text, a unigram, a bigram and a trigram were computed [Lan98]. These three terms are based on the *n-gram* methods, which is defined as the procedure that splits a text in *n* size chunks, here one, two and three words, so that *n* sub-sequentially following words in a text a put together [Lan98]. For example, for the sentence *hi my name is Tim*, the unigram is every single word, the bigrams are *hi my*, *my name*, *name is* and *is Tim*. Lastly, also the trigram, which is *hi my name is*, *my name is* and *name is Tim*, was computed. The assumption was, that a novel keyword consists of a maximum of three words, because there is also no word beyond three words in the predefined keyword list from Appendix B.
Every episode with an average of 5,000 words was used to compute the unigrams, bigrams and trigrams out of it. Each chunk was inferred from these approximately 13,000 one-to-three word combinations with the Vosk episode-level topic model. If the text chunk was classified as being the *ai* topic, the word was saved in a separate file. After filtering duplicates and nonsense chunks, the following new keywords were identified:

- dimension,
- multimodal,

- digital medicine,
- bioengineering.

5.4 Sentiment Analysis

5.4.1 Sentiment Development over Time

The sentiment analysis was conducted to visualize the sentiment of each podcast towards each of the found topics. The sentiment transformer that was used from the Hugging Face platform [Huga], provides a value from 0 to 1 and a label (either positive or negative), where the value between 0 and 1 represents the likelihood of the label. The sentiment score for the figures in this section were computed as a relative share between positive and negative within the range of 0 to 1, where 0 represents a total negative and 1 represents a total positive sentiment (Section 4.4.3). Figure 5.11 shows the average sentiment of every podcast for each of the topics. The darker the green shade of a field in the heatmap becomes, the more positive the sentiment is in the according field. In summary, the podcasts have a positive or very positive sentiment towards most of the topics, such as *startup acceleration*, *healthcare innovation* or *electronic health records*. Only in the years 2015 and 2016 a neutral sentiment is present in the lower part of the y-axis. It shall be recalled again that Figure 5.1 illustrated that there was a low amount of total episodes in the years 2015 and 2016. Not all podcasts have started yet in these two years, as can be read from Table 4.1. The contribution of single podcasts to the average sentiment shown in Figure 5.11 will be analyzed in more depth. For this analysis, two random podcasts with both an overall positive and an overall negative sentiment were chosen.

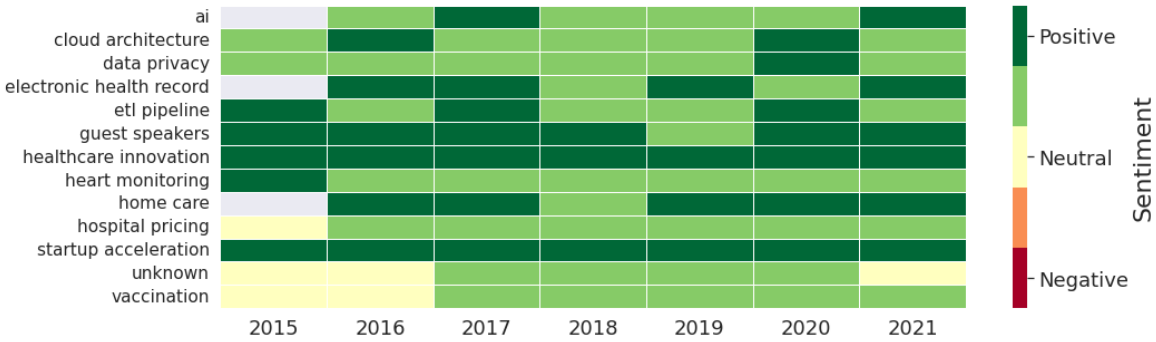


Figure 5.11: Yearly averaged sentiments of the topics for all 29 healthcare podcasts.

Figure 5.12 shows a podcast with very positive sentiment on average, namely the podcast *HealthcareChangers*. This podcast started in the year 2017 and, according to its description, claims to talk about:

> *"Real people's stories at a personal level and dive deep into specific health care topics, from medical billing and new technologies to palliative care and everything in between"* [Cam17].

It can be seen, that the topics *startup acceleration*, *heart monitoring* and *healthcare innovation* occur frequently with an overall very positive sentiment. Comparing these three topics with its podcast description, they appear to match with it.

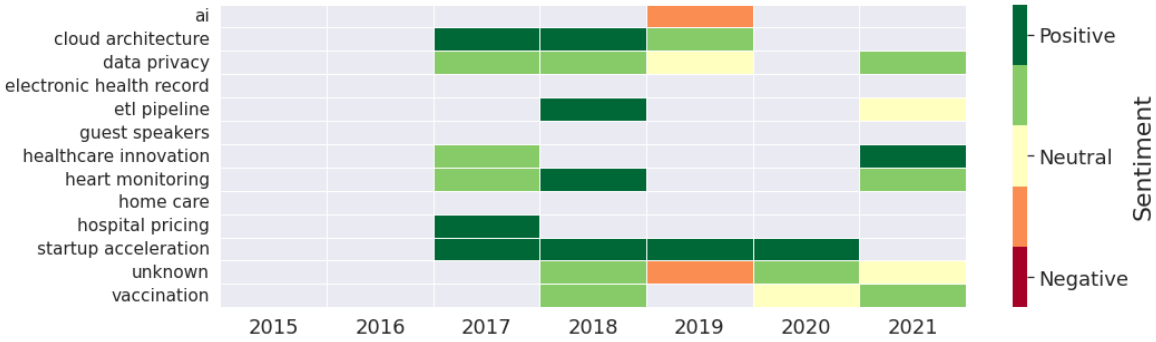


Figure 5.12: Yearly sentiments of the topics from the podcast *HealthChangers* [Cam17]

Another more negative example is the podcast *Healthcare Triage Podcast*. The hosts of the podcast started to produce episodes in 2015. This podcast's description claims to talk about:

> *"explain healthcare policy, medical research, and answers a lot of other questions you may have about medicine, health, and healthcare"* [Dr 15].

Figure 5.13 reveals, that in the *Healthcare Triage Podcast* there are many conversations about *data privacy* and about *vaccination*, even before the COVID-19 pandemic occurred [Pla20]. From 2015 to 2017, the overall sentiment of this podcast was very negative. A yearly change towards more positive sentiments within the same topics can be noticed. The term *healthcare policy* from the description seems to match with the occurring topic *data privacy*. Furthermore, among all topics, the topic *vaccination* especially in combination with COVID-19 seems to match the medical research topic from the podcast's description.

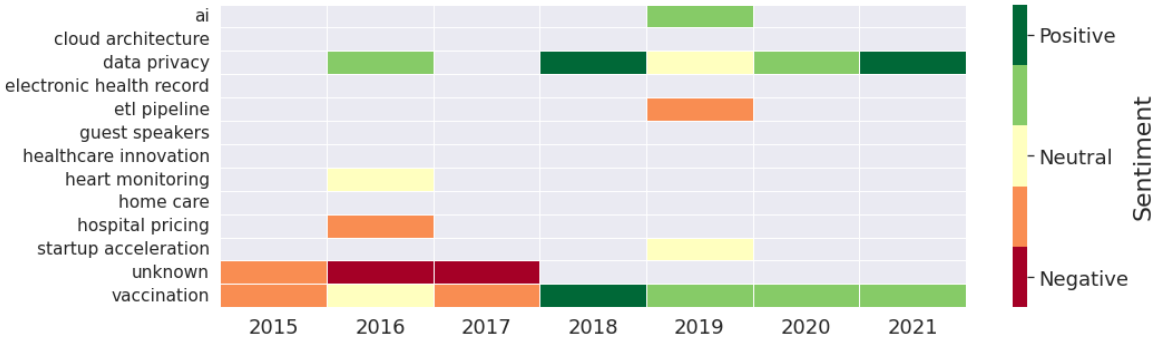


Figure 5.13: Yearly sentiments of the topics from the podcast *Healthcare Triage Podcast* [Dr 15]

5.4.2 Emotional Analysis

Another more dedicated approach was also tested out. The Hugging Face [Huga] platform also provides emotional classifiers. This approach was inspired by [Fas17] with their emotion analysis of New York Times articles. For this purpose, the model *j-hartmann/emotion-english-distilroberta-base* from Hugging Face was used. It showed, that even if in theory eight potential emotions could be classified, most of the classifications were *fear* and *joy*. This result is similar to being positive or negative with the standard sentiment analysis. For this reason, further research was omitted because no new information could be gained through this approach.

Chapter 6

Discussion

6.1 Data Set

The data set is a crucial part of this thesis. It is the basis of all data analysis steps and corresponding plots resulting from the analysis. For that reason, great care was taken to apply all evaluations homogeneously. The crawling pipeline from Do and colleagues [Phi21] was used to download all podcasts and metadata. Due to the nature of RSS-Feeds, some links or the format of how the episodes are stored changes over time. For that reason, one podcast (*The Big Unlock*) did not allow to crawl the metadata anymore. This could happen again when the pipeline is used in the future. Adaptations to the code are necessary when significant changes of the RSS-Feeds happen.

Due to the lack of research on large texts and topic clustering, the 3,449 episode transcriptions were split into smaller chunks. Each episode transcription was investigated for the appearing keywords from the keyword list. For each keyword in the episode's text, 200 words were taken into the chunk before and after each keyword. This procedure resulted in a total length of 401 words per chunk. However, this length of 401 words was decided based on Figure 4.4 and the length of words from common social media posts (Twitter Tweet length of 280 [Twi]). The assumption was, that with an average occurrence of ten keywords in an average episode text with a length of 5,000 words, a chunk with 401 words would not overlap much with other chunks found in the same episode text. The overlap for the length of 200 was 36% and a total of 40% of the entire episode text was used on average. Different results could have been found if the length of 401 words per chunk had been longer or shorter. Research remains

to be conducted about the topic model results of different lengths of the text chunks. Due to the length of 280 words per Twitter Tweet and the percentage of overlapping and preserved words, it is recommended to experiment with a smaller chunk size between 201 and 301 total words, so a length of prior and post words between [100, 150] for k.
The podcast *Outcomes Rocket* (OR) in Figure 5.2 produced four times more episodes than *PopHealth Week* (PW), the podcast with the second most episodes. Even though *Outcomes Rocket* started in 2017, the podcast creators produce episodes almost daily. This imbalance embraces a huge data imbalance that must be taken care of. The found topics could depend too much on this specific podcast.

6.2 Speech-to-Text APIs

The first research question for this thesis was defined in Section 1.2. The desired goal was to determine whether Vosk or DeepSpeech performs better for the transcription process of podcast audio files. The transcription process should be accessible to everyone who wants to recreate or elaborate on this thesis's results. For this purpose, open-source APIs were used. Do and colleagues [Phi21] already evaluated the WER of Vosk and DeepSpeech. The team picked one episode from each of the 30 podcasts and selected 30 seconds from the beginning, the middle, and the end of each episode. This approach results in 90 snippets (30 podcasts times three snippets) with 30 seconds per snippet transcribed by hand and then evaluated with the WER metric. Nevertheless, this method is problematic since many podcasts have an intro or outro melody. A bias due to an accidental tracking of random words from the intro or outro jingles is almost inevitable. For this reason, the more robust approach of *randomly* selecting three times 30-second snippets in each of the 30 episodes was additionally evaluated. None of the three snippets could accidentally overlap to guarantee a robust evaluation. It showed, that both APIs perform almost equally in both evaluations. The average mean WER of both APIs differs only by a value of 2.5 for the second evaluation. However, the boxplot for the second evaluation (Figure 5.3) reveals that DeepSpeech has a wider standard deviation. This deviation appears to be present because of the *wer_deepspeech_2* snippets, which have the worst WER among all averaged snippets. There are multiple reasons why the performance of this particular snippet is so poor. The most reasonable is due to the randomness of each 30-second snippet. Since the snippets were taken at a random spot in the audio file, it could be the case that a spoken word is divided in two at the beginning or the end of a snippet, and

the word cannot be heard and transcribed correctly anymore. This could have led to a wrong word order due to misspelling and a skew of the WER metric.
Another finding that can not be explained is the different lengths of both API keyword data sets. For Vosk, the keyword data set has a length of 35,816. On average in all 3,449 transcribed episodes around ten keywords per episode have been found. On the other hand, for the 3,449 transcribed episodes with DeepSpeech, only 14,487 keywords in total have been found. Great care was given to apply the same cleaning and mapping approaches for both raw text transcriptions. Nevertheless, in the DeepSpeech transcriptions, approximately only half of the buzzwords have been found compared to the Vosk transcriptions. A deeper investigation is necessary to determine why this is the case.
The acceleration of the GPU for the API DeepSpeech turned out to be more complex than expected, because of the exact CUDA [NVIa] version of 10.1 that was required. Contrary to that, it worked flawlessly for the API Vosk to transcribe the *MP3* files into text files with GPU acceleration. Considering that Vosk's average WER is worse for Vosk than DeepSpeech, the transcriptions from Vosk contain twice the number of keywords from the keyword list. Furthermore, the GPU acceleration proved easier for Vosk than for DeepSpeech. The research leads to the overall recommendation of Vosk instead of DeepSpeech for further research.

6.3 Topic Clustering

The second research question from Section 1.2 dealt with the usage of topic clustering to identify new insights for trend detection. At the beginning of this research, it was unclear which topic clustering algorithm to use. After conducting the literature research of state-of-the-art topic models, neural topic clustering approaches were identified as the currently best performing topic models [Vay20]. However, there exist a bright variety of different neural topic models. Therefore, more literature research was conducted to identify which neural topic modeling approach shall be used and which package from Github [Gita] is suitable to fulfill the goals of this thesis. The Hugging Face [Huga] open-source platform for transformer models was the ideal solution due to the effortless way of implementing the models. Another reason was that the Hugging Face platform is open-source, aligning with the goals of using open-source technologies for this thesis. In order to evaluate the objective performance of the topic models, it was desired to compute metrics that evaluate the topic models' performance. The *Coherence* and the *NPMI* metrics are common metrics to evaluate the performance of

topic models (Section 4.1.2). Since the OCTIS [Ter21] package provides the computation of the metrics off-the-shelf, OCTIS was identified as the neural topic modeling package of choice. Nevertheless, metrics like the *logs-ratio* or the *Ranked-Biased-Overlap* could have led to other decisions about choosing the neural topic modeling package.

The topics over time found by the neural topic clustering with the BERT model *all-mpnet-base-v2* were novel and consistent with trends according to the given time frame. For example, topics about *vaccination* occurred during the COVID-19 pandemic. The keyword list includes many relevant topics, because 23 out of the 102 were found in the 3,449 transcribed episodes. The found topics included words from the keyword list and new words with similar meanings but not included in the list. Nevertheless, the new words are not bound to be technology-related. This finding assumes that particular terms not defined in the keyword list are almost not occurring in the podcast speeches. Therefore, topic clustering can identify words with similar meanings from the found topic strings to those from the keyword list. However, performing topic clustering itself does not discover novel words not already present in the keyword list. Furthermore, since only keywords from the keyword list were used to create the text chunks, the topic clustering could be biased towards these topics.

Apart from the topic model topic words itself, the topic model was used together with *n-grams* to identify novel keywords within the raw Vosk transcribed episode-level texts. Using the topic model in such a way, novel keywords, such as *dimension* or *multimodal*, were identified. However, suppose these new keywords were added to the keyword list and the entire topic modeling process was repeated. In that case, it is unclear how the new topic model will be biased since only words that the topic model itself has identified were added to the list. Nevertheless, the possibility of identifying novel keywords within the domain of a specific topic like *ai* was accomplished.

6.3.1 Subtopic Clustering

It is worth mentioning that another more experimental approach was conducted to identify novel keywords with topic clustering. Each of the 35,816 snippets with a length of 401 words was predicted with one topic to create the plots from Chapter 5. Fourteen distinct topics were found, whereas one topic was identified to be AI-related. A new data set of the length of 2,206 was created that only consisted of data from the Vosk keyword data set that was assigned to the topic AI. This pure AI-related keyword data set was used as input to train a new topic

clustering model, which should therefore find sub-topics of the topic AI. The identified topics can be found in Appendix C. One of the topics could care about *early detection* and another one about *data collection*, but further research needs to be conducted on how this information could be used for further trend detection.

6.4 Sentiment Analysis

The last research question from Section 1.2 shall answer whether sentiment analysis could reveal new insights for trend detection. The sentiment analysis was used to identify the sentiment of the speakers towards the found topics in the transcriptions. Generally, sentiment analysis is a valuable tool to visualize the sentiment over time towards specific topics and reveal a trend of hype or fear towards topics. After the topic clustering was performed, the sentiment was computed for every of the 29 podcasts towards each of the 14 topics. Because podcasts are supposedly more positive than negative, it is still interesting to see podcasts that talk especially positively or negatively about specific occurring topics. If successful, a podcast selection can filter podcasts with undesired sentiments. The sentiment could further reveal if podcast hosts and guest speakers fear or hype the development of specified topics. Nevertheless, on average a very positive sentiment was computed for most of the topics every year. However, no correlation was identified between the found topics and the sentiment. Sentiment analysis is a valuable tool for filtering by personal preference but not for identifying new keywords or correlations between topics.
The second approach that was applied is the emotional analysis. Since no new insights were found, further research was omitted because all emotions could be broken down into positive and negative sentiments. However, if smaller text chunks were used, more diverse emotions could emerge. Furthermore, new transformer models could be uploaded to the Hugging Face [Huga] platform that are more suitable for the podcast domain.

Chapter 7

Conclusion, Limitations and Outlook

7.1 Conclusion

This work was built upon the pipeline created by Do and colleagues [Phi21]. Over 4,000 podcast episodes from 30 handpicked healthcare podcasts were downloaded with the pipeline, from which 3,449 episodes with a total length of 60,000 hours of speech fulfilled all criteria for further research. Speech-to-text transcriptions and evaluations were conducted on the open-source APIs Vosk and DeepSpeech, with a clear recommendation for the API Vosk. The keyword list provided by Do and colleagues [Phi21] was refurbished and extended with novel keywords from well-known machine learning glossaries. The Vosk transcriptions and the keyword list were used to build the Vosk keyword-level data set of 35,816 text chunks with a length of 401 words each, where the found keyword is in the middle of the chunk. Topic clustering on this keyword-level data set revealed, that trend prediction towards specific topics is possible. Identifying new keywords by selecting only the AI-related text chunks from the keyword-level data set and performing topic clustering again on these subtopics showed interesting results. Also, the keyword search with the *n-grams* identified novel keywords. In total four new keywords, such as *dimension* or *multimodal*, have been added to the keyword list. The sentiment analysis on the predicted topics appeared to work well and correctly identified sentiment trends towards specified topics. Podcasts can be distinguished by filtering the sentiments for each topic. In summary, topic clustering and sentiment analysis are helpful approaches to perform trend detection for podcast data. Future research with more data could allow using subtopic clustering in more depth. Furthermore, introducing speaker diarization

could help to better separate guest speakers from podcast hosts and potentially gain more insights out of the data [Par22].

7.2 Limitations

Several limiting factors occurred during this thesis. The first one was the mandatory selection of an open-source API. However, Do and his team [Phi21] showed, that Microsoft's Azure [Mic] performed the best among all APIs, but it is a commercialized transcriptor. Another limiting factor was the data itself. Even though only 30 podcasts have been selected (from which 29 were used later), the transcription process with Vosk took approximately two and a half weeks of pure computation time. This thesis could not have been accomplished if more podcasts had been chosen. Furthermore, finding the ideal number of clusters can also be improved. A model was trained with the step size of two from [10, 40] clusters (Figure 4.5). If a step size of one was chosen, the optimal clusters could be even more accurate, because not only even, but also odd numbers can be checked.

Another major problem can be seen in Figure 5.1 because the podcast *Outcomes Rocket* has four times more episodes than the podcast with the second most episodes. This could lead to a potential data imbalance due to a skewed database toward topics discussed in the *Outcomes Rocket* podcast. For this reason, it is recommended that rather more podcasts but fewer episodes per podcast are included in the data set.

The optimization step of the topic clustering was minimal due to the required computational power. The length of 35,816 training samples with 401 words each requires a lot of Random Access Memory. Even though the topic clustering was performed on a machine with 64 Gigabyte of RAM installed, the required memory exceeded the limits of the installed memory after each optimization step. Therefore, only one step of optimization could be accomplished at a time. More optimization steps could improve the overall performance of the topic clustering algorithm. If more podcast data had been available, the topic clustering optimal number of topics and optimization would demand even more resources.

The selected chunk size of 200 words prior and post to the found keywords also limits the conducted research. Varying chunk sizes could produce different results and different topics could have been found. For this reason, different chunk sizes must be tried out to conduct topic clustering more robustly as shown in Figure 4.4.

7.3 Outlook

Future work could include increasing the number of podcasts and episodes. As explained in Section 7.2, more data requires more resources, but this seems like the easiest way to find more keywords from the keyword list.
The BERT model itself has a significant impact on the feature extraction of the text and, therefore, of the topics found in the topic model. The *all-mpnet-base-v2* was used as BERT model. If multilingual models or models for specific medicine-related purposed were used, the found topics could also differ substantially.
Since 29 out of the 102 keywords were found in the 3,449 episodes, the topic models' output was used to find more topics. The models' search process took very long because it was performed on the raw text. If a Named Entity Recognition (NER) was used to filter out entities and only check if they match with the *ai* topic, a lot of computation time could be saved if the same output is produced. Furthermore, since 14 topics have been identified, a keyword search for every possible topic could also be conducted.
The sentiment analysis has shown that hype and fear of podcast speakers can be identified. Nevertheless, it is expected that guest speakers appear in podcast episodes. The text used for this thesis is regarded as one speaker, even though the speech was recorded by multiple people having a conversation. The introduction of speaker diarization could enable new insights by dividing the transcribed text into podcast hosts and guest speakers [Par22].
The idea of subtopic clustering was introduced in Section 6.3.1. If more data was available, an improved subtopic clustering could reveal subtopics that can be used to improve the trend detection and identify novel topics and subtopics.
Finally, not only using sentiment analysis but using a broader range of emotions could reveal more insights about trends. Future research could use more advanced models for topic clustering or emotion analysis. There is continuous improvement of the models on the Hugging Face platform [Huga].

7.4 Acknowledgement

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Appendix A

Acronyms

AI Artificial Intelligence

ALBERT A Lite BERT for Self-supervised Learning

API Application Programming Interface

ASR Automatic Speech Recognition

BERT Bidirectional Encoder Representations from Transformers

CPU Central Processing Unit

CUDA Compute Unified Device Architecture

DL Deep Learning

GPU Graphics Processing Units

LDA Latent Dirichlet Analysis

ML Machine Learning

NLP Natural Language Processing

OCTIS Optimizing and Comparing Topic Models is Simple

NER Named Entity Recognition

NPMI Normalized Pointwise Mutual Information

RAM Random Access Memory

RSS Really Simple Syndication

WER Word Error Rate

XLM Cross-lingual Language Model Pretraining

Appendix B

Keyword List

English Buzzwords		
Algorithm	Digital Ethics	Neuromorphic Hardware
Analytics	Digital Transformation	Neuron
Api	Dynamic Programming	Node
Artificial Intelligence	Encoder	Noise
Artificial Intelligence Cloud Service	Epoch	Normalization
Artificial Intelligence Governance	Error Rate	Optical Character Recognition
Augmented Reality	Expert Systems	Outlier
Bayesian Network	Explainable Artificial Intelligence	Parameter
Bias	Extrapolation	Perceptron
Big Data	Fuzzy Logic	Predicate Calculus
Black Box	Game Theory	Predictive Analytics
Bots	Gaussian Distribution	Principal Component Analysis
Business Analytics	Generalized Linear Model	Python
Chatbot	Generative Adversarial Network	Quantization
Cloud	Generator	Quantum Computing
Clustering	Genetic Algorithm	Reinforcement Learning
Cognitive Computing	Genetic Programming	Responsible Artificial Intelligence
Cognitive Science	Gesture Recognition	Robotic Process Automation
Computer Vision	Hashing	Self Supervised Learning
Convolution	Heuristic	Signal Processing
Correlation	Image Recognition	Smart Machines
Covariance	Induction	Software
Cross-validation	Inference	Speech Analytics
Cybernetics	Information Retrieval	Token
Dark Data	Information Theory	Transfer Learning
Data	Internet Of Things	Transparency
Data Labeling	Knowledge Representation	Tuning
Data Mining	Label	Turing Test
Decision Intelligence	Machine Intelligence	Variance
Decision Tree	Machine Learning	Verification
Decoder	Machine Translation	Video Analytics
Deep Learning	Metric	Virtual Reality
Deep Neural Networks	Natural Language Processing	Web 2.0
Descriptive Analytics	Neural Network	Web 3.0

Table B.1: Buzzword list in English containing 102 buzzwords used for the data analysis.

Appendix C

AI Subtopics

	Sign Language	Early Detection	Data Collection	Measuring Device	Unknown
1	sort	model	noise	level	reception
2	normal	early	field	dial	know
3	sign	kind	accuracy	recognition	dealership
4	code	software	generate	device	vertically
5	nodule	system	box	reception	odom
6	eg	patient	prediction	measure	expandability
7	actually	gene	live	disperse	spawn
8	product	probably	collect	detection	eightth
9	device	add	deal	processing	change
10	tie	network	produce	potus	datum

Table C.1: AI Subtopic list containing ten corresponding words for each subtopic.

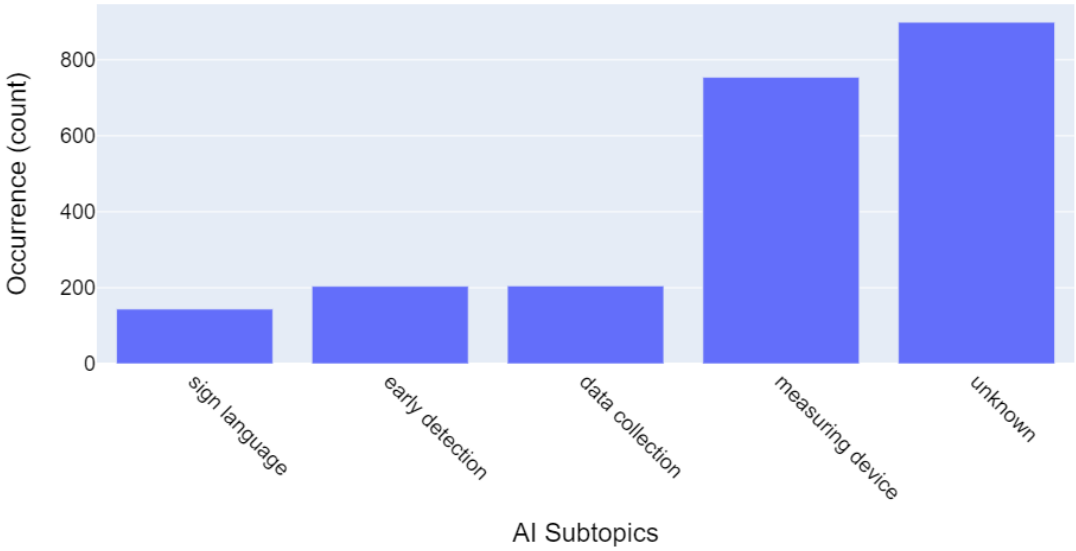

Figure C.1: Total occurrences of the five subtopics.

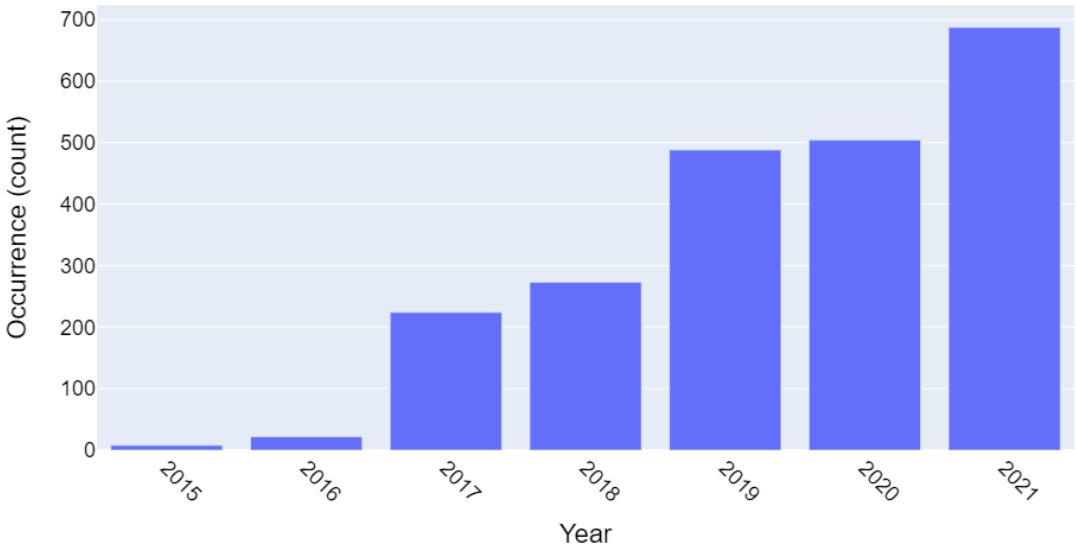

Figure C.2: Yearly occurrences of the AI subtopics.

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